

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Interconnects (Part 2)

Instructor: Prof. Joshua San Miguel

Some slides adopted from Natalie Enright Jerger

Last Lecture

- Relaxed consistency examples
- Interconnects
- Network diameter
- Channel load

This Lecture

- Bisection bandwidth
- Path diversity
- Direct topologies
- Indirect topologies

Network-On-Chip – Evaluation

Metrics for evaluating NoC performance and cost/complexity:

- Degree
- Hop count
- Network diameter
- Maximum channel load
- Bisection bandwidth
- Path diversity

Network-On-Chip – Bisection Bandwidth

Network Cut:

Partition that splits all nodes into two disjoint sets

Network Bisection:

Cut with the same number of nodes in each set (or nearly the same)

Bisection Bandwidth:

Minimum cross-cut bandwidth among all bisections

- Proxy for estimating cost (i.e., higher bisection bandwidth implies more global wiring)
- Affected by topology, routing and flow control

Bisection Traffic:

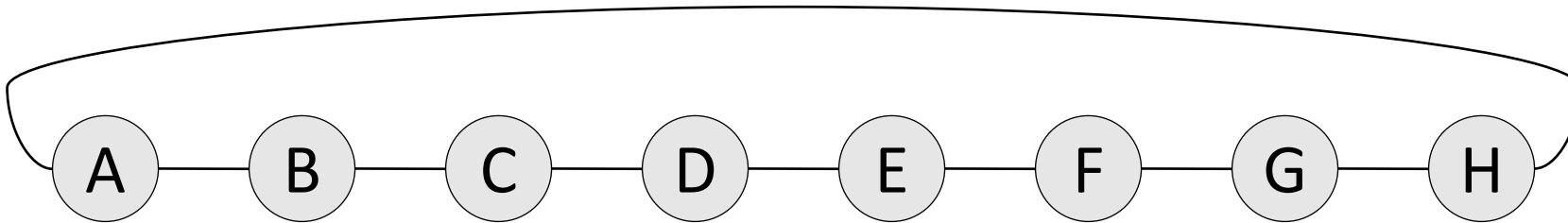
Amount of traffic crossing minimum bisection

- Proxy for estimating saturation throughput

Network-On-Chip – Bisection Bandwidth

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count

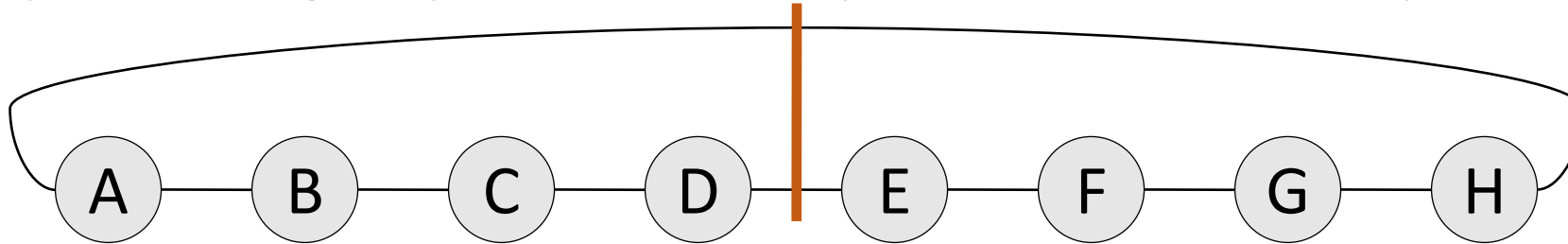


What is the bisection bandwidth?

Network-On-Chip – Bisection Bandwidth

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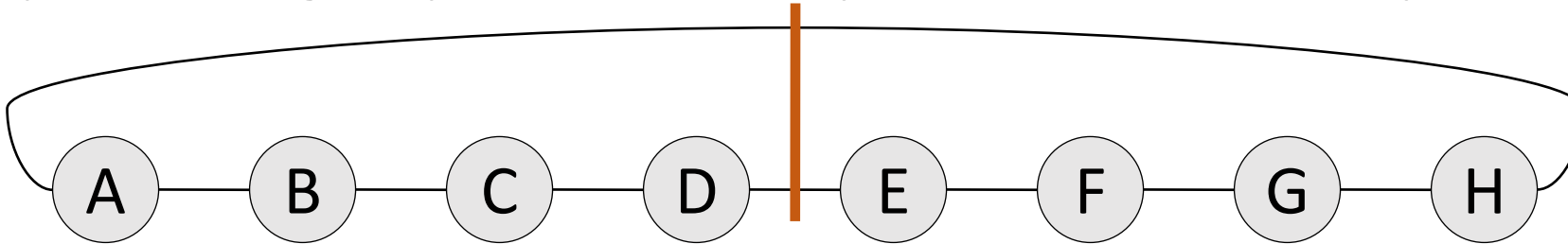
What is the bisection bandwidth?

4 unidirectional links

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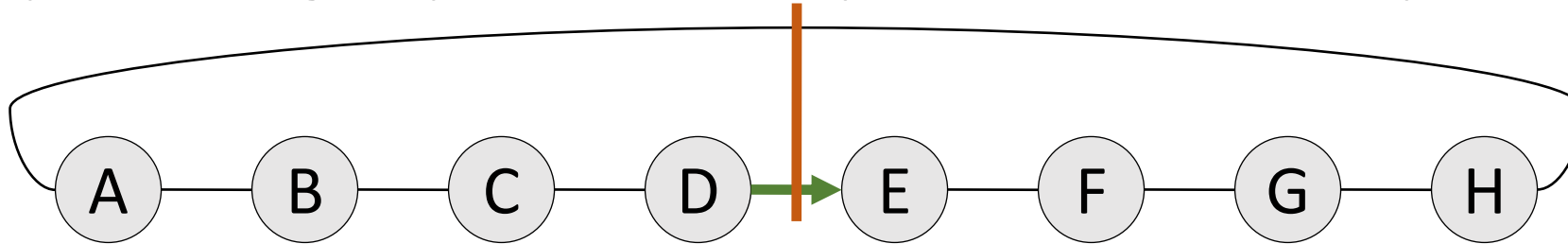


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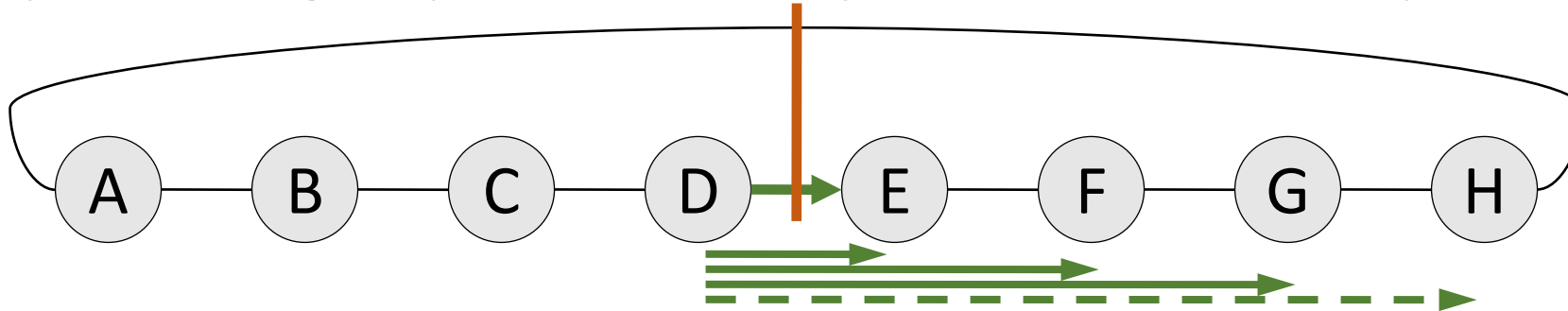


At link from D to E:

Network-On-Chip – Bisection Traffic

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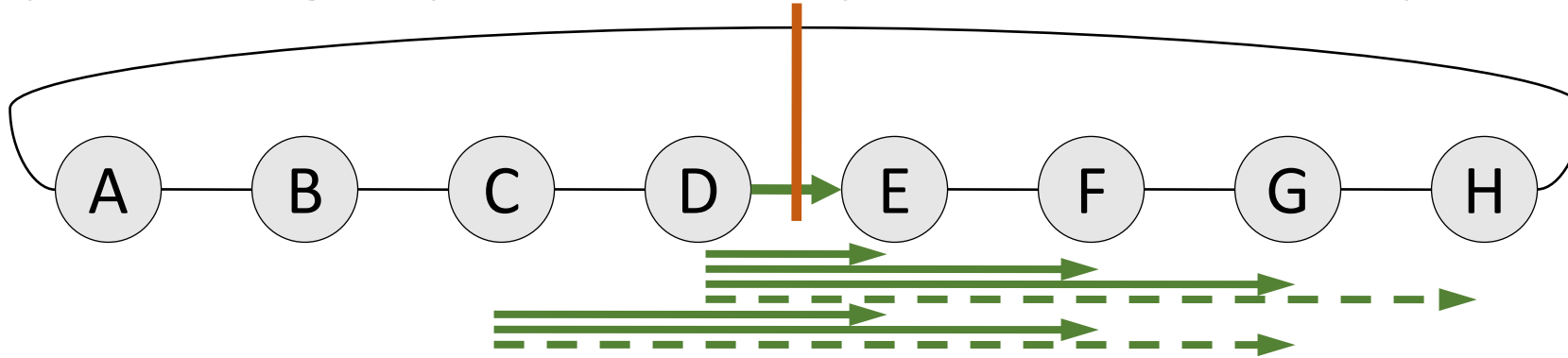
At link from D to E:

- D sends $1/8$ of its traffic to E, F, G and $1/16$ of its traffic to H (since 2 possible shortest paths to H).

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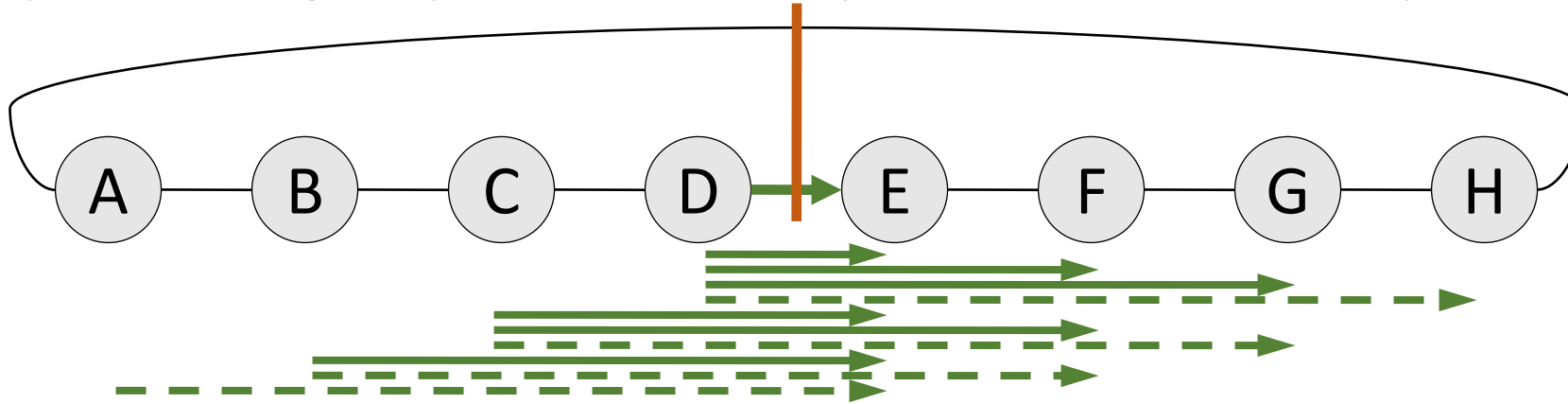
At link from D to E:

- D sends $\frac{1}{8}$ of its traffic to E, F, G and $\frac{1}{16}$ of its traffic to H (since 2 possible shortest paths to H).
- C sends $\frac{1}{8}$ of its traffic to E, F and $\frac{1}{16}$ of its traffic to G.

Network-On-Chip – Bisection Traffic

e.g., Assume:

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At link from D to E:

- D sends $1/8$ of its traffic to E, F, G and $1/16$ of its traffic to H (since 2 possible shortest paths to H).
- C sends $1/8$ of its traffic to E, F and $1/16$ of its traffic to G.
- B sends $1/8$ of its traffic to E and $1/16$ to F, and A sends $1/16$ to E.

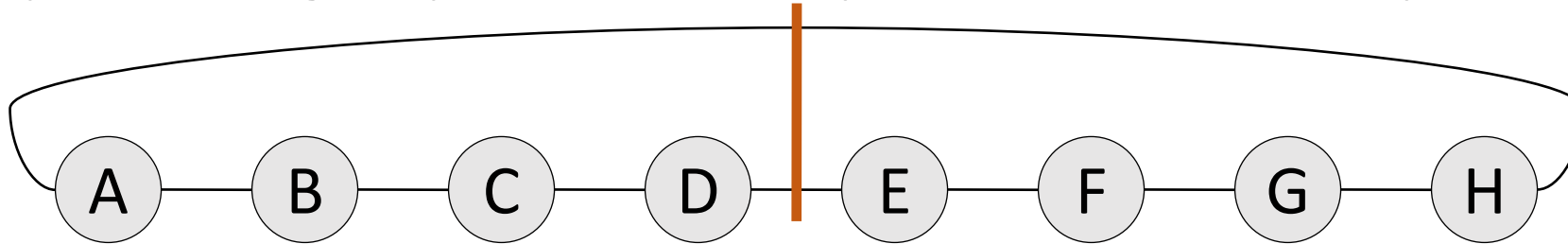
Therefore, channel load of link from D to E is 1 packet per cycle.



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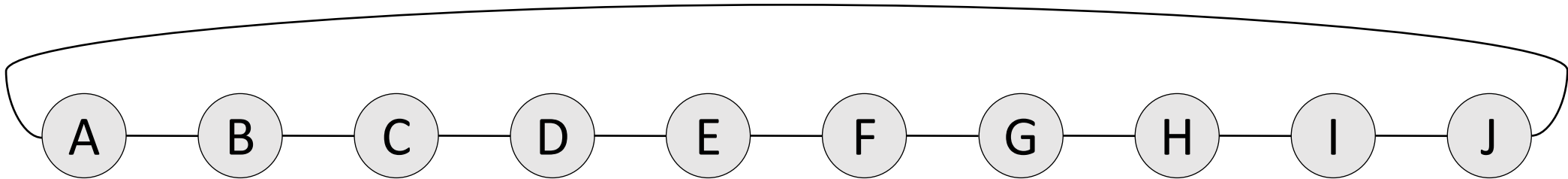


Channel load of link from D to E is 1 packet per cycle.
Channel load of link from E to D is 1 packet per cycle.
Channel load of link from A to H is 1 packet per cycle.
Channel load of link from H to A is 1 packet per cycle.
Therefore, bisection traffic is 4 packets per cycle.

Network-On-Chip – Bisection Bandwidth

e.g., Assume:

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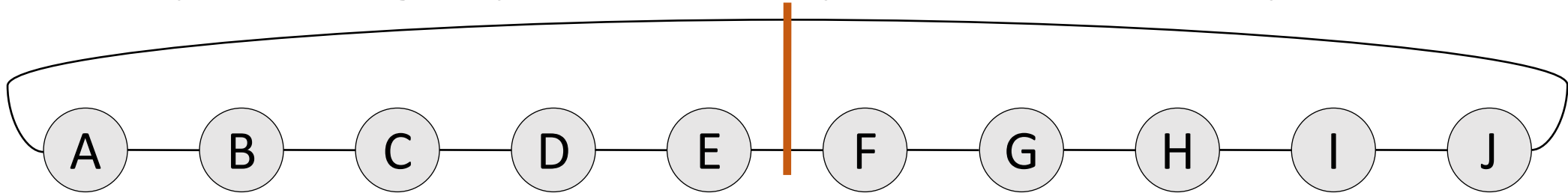


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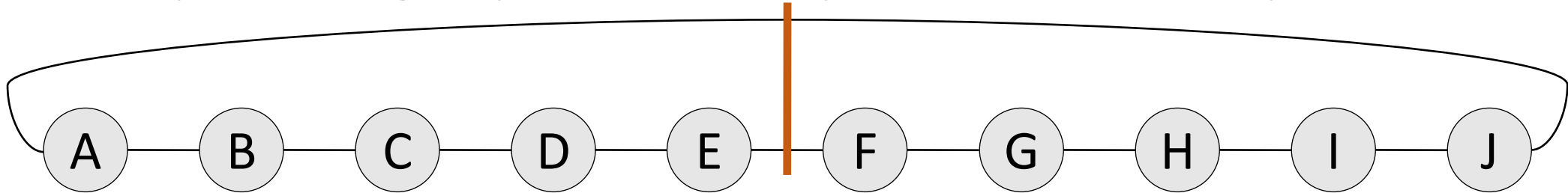
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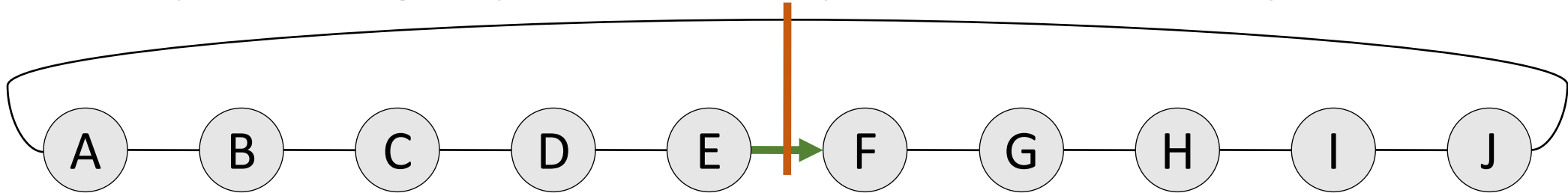


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Network-On-Chip – Bisection Traffic

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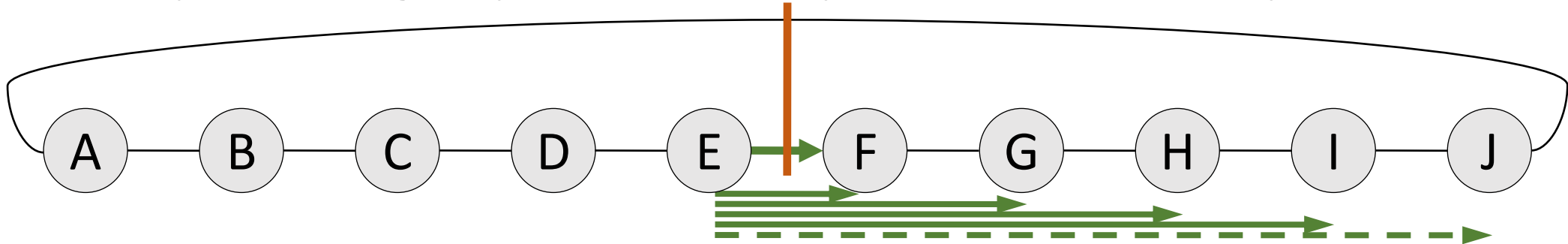


At link from E to F:

Network-On-Chip – Bisection Traffic

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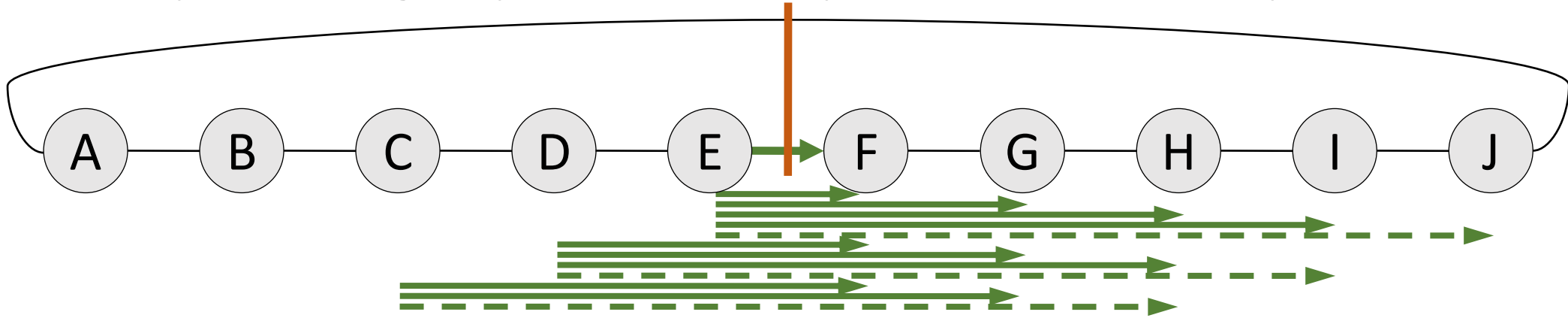
At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.

Network-On-Chip – Bisection Traffic

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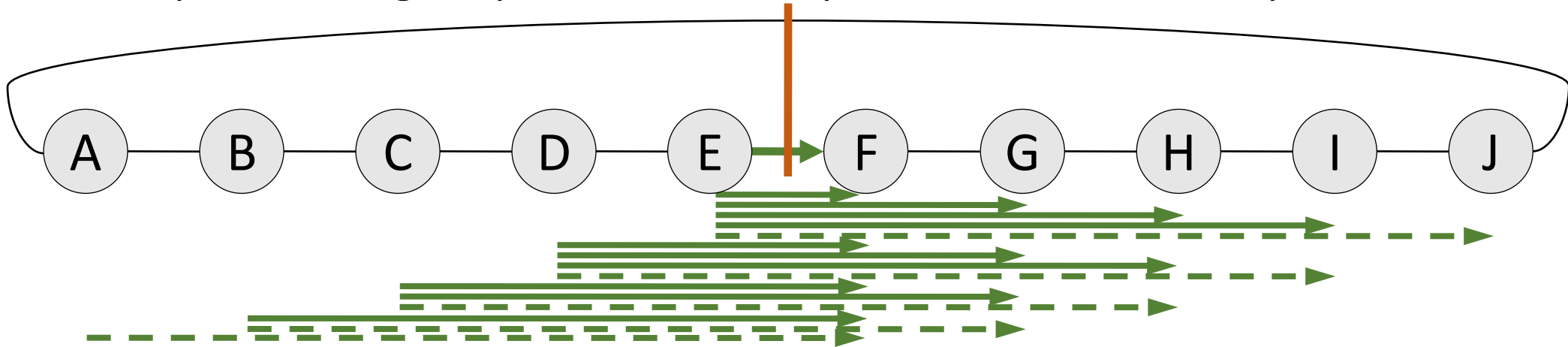
At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.
- D sends $1/10$ of its traffic to F, G, H and $1/20$ to I. C sends $1/10$ to F, G and $1/20$ to H.

Network-On-Chip – Bisection Traffic

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
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At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.
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- B sends $1/10$ to F and $1/20$ to G. A sends $1/20$ to F.

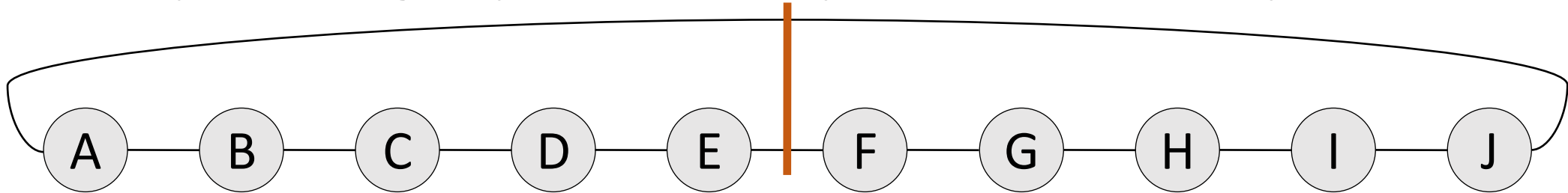
Therefore, channel load of link from E to F is 1.25 packets per cycle.



Network-On-Chip – Bisection Traffic

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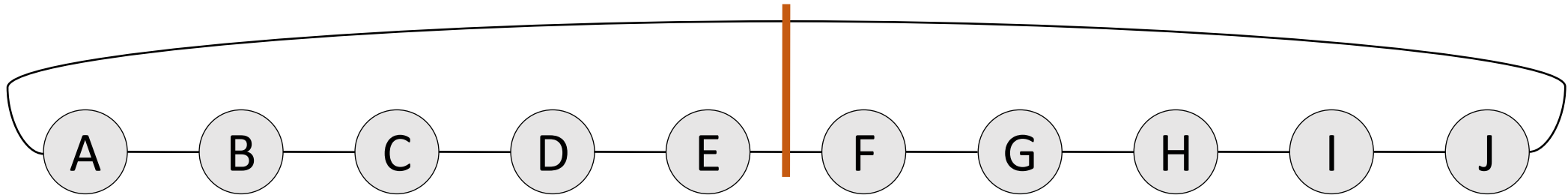


Channel load of link from E to F is 1.25 packets per cycle.
Channel load of link from F to E is 1.25 packets per cycle.
Channel load of link from A to J is 1.25 packets per cycle.
Channel load of link from J to A is 1.25 packets per cycle.
Therefore, bisection traffic is 5 packets per cycle.

Network-On-Chip – Bisection Traffic

Since bisection bandwidth is only 4 packets per cycle:

- Then the network saturates at 4/5 of the max injection bandwidth (i.e., all nodes inject 4 packets per 5 cycles)



Channel load of link from E to F is 1.25 packets per cycle.
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Channel load of link from A to J is 1.25 packets per cycle.
Channel load of link from J to A is 1.25 packets per cycle.
Therefore, bisection traffic is 5 packets per cycle.

Network-On-Chip – Path Diversity

Path Diversity:

Number of possible paths between each source-destination pair

- Proxy for estimating fault tolerance and load balancing
- Affected by topology and routing

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Network-On-Chip – Path Diversity

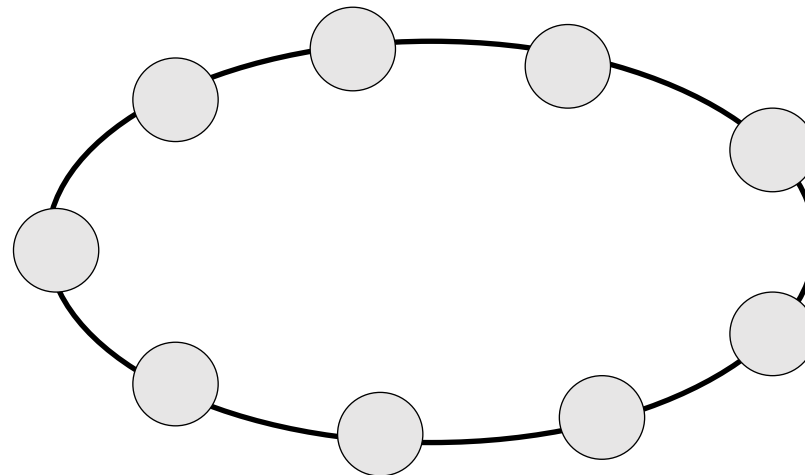
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Path diversity?



Network-On-Chip – Path Diversity

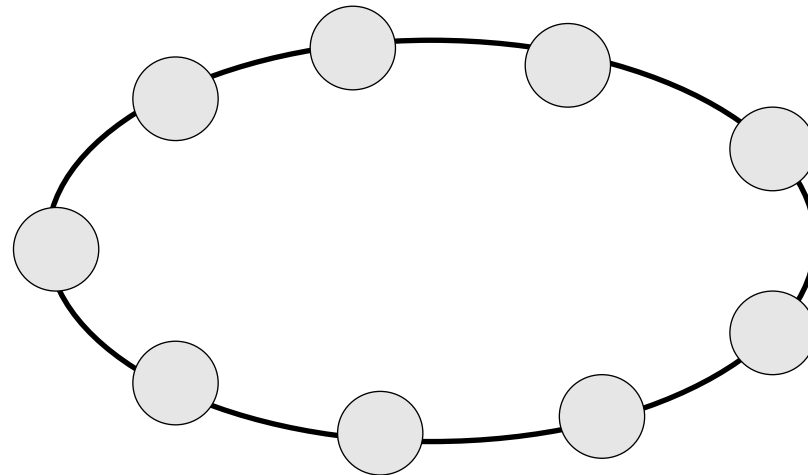
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Path diversity? 1

Network-On-Chip – Path Diversity

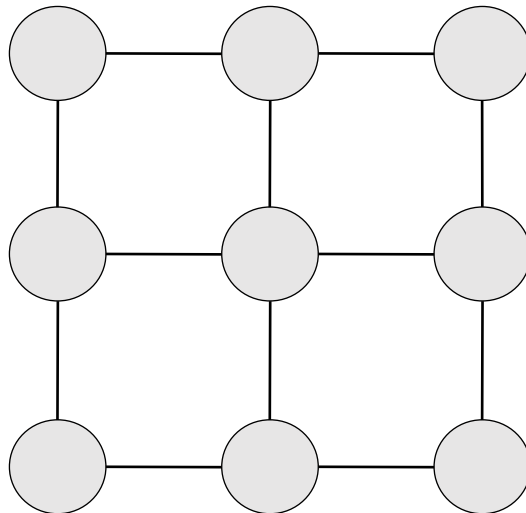
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Min path diversity?
Max path diversity?

Network-On-Chip – Path Diversity

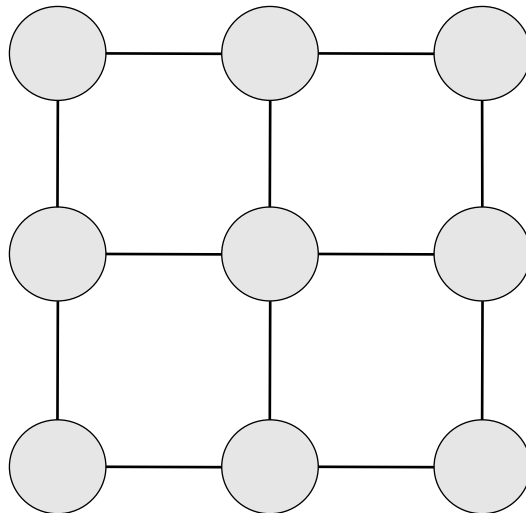
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Min path diversity? 1

Max path diversity?

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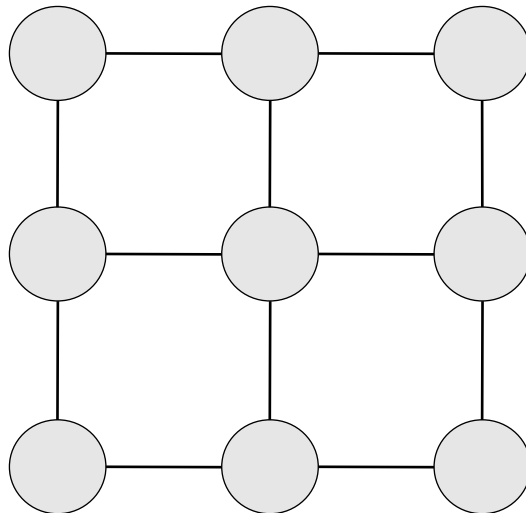
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Min path diversity? 1

Max path diversity? 6

Network-On-Chip – Path Diversity

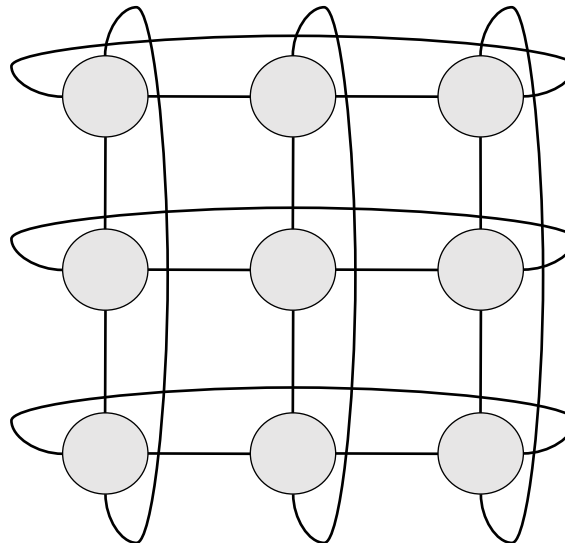
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Max path diversity?

Network-On-Chip – Path Diversity

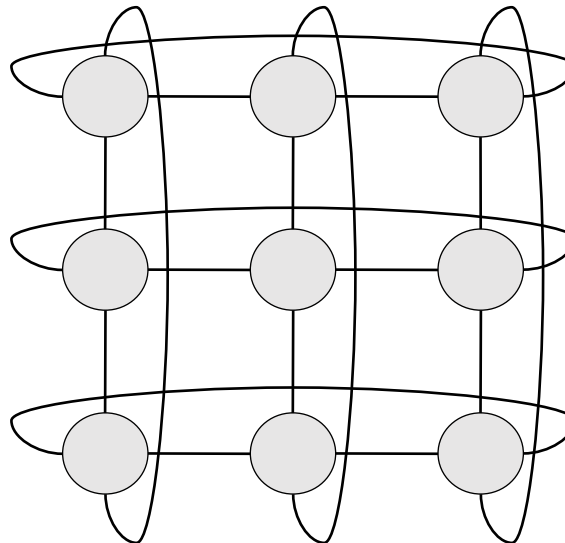
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Max path diversity? 2

Network-On-Chip – Path Diversity

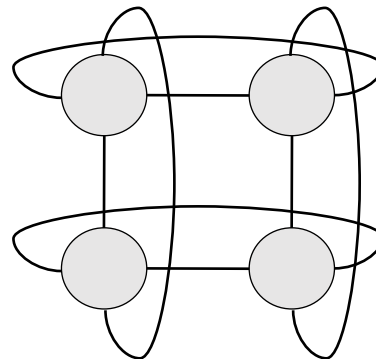
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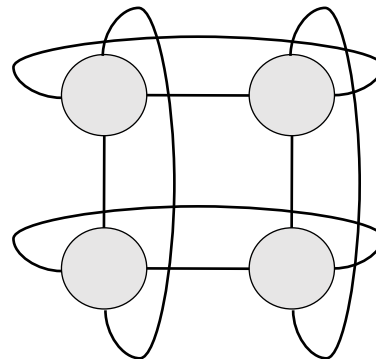
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Max path diversity? 8



Network-On-Chip

Key design decisions:

Topology

- Where to place routers/nodes and how to link them together

Routing

- Which set of links to traverse between source-destination pairs

Flow Control

- What granularity to move messages through links and routers

Router Microarchitecture

- How to pipeline and arbitrate incoming messages in routers

Topology

Arrangement of nodes and links (or channels) in network

- Analogous to road map
- Often first step in NoC design
- Impacts network latency and energy
- Impacts layout complexity and router complexity

Assumptions, for now:

- Ideal routing/routers (i.e., perfect load balancing among links)
- Ideal flow control (i.e., no idle cycles on any links)

Topology

Bus:

- Single channel shared by all nodes
- Effective broadcast medium

Crossbar:

- Connects x inputs to y outputs without intermediate hops
- High throughput for distinct input-output flows
- Generally used to connect ports within routers

Router-based:

- Packets traverse through links and routers (multi-hop paths)
- Each router operates independently to determine where the packet goes next
- Main focus of current NoC research

Topology – Direct and Indirect

Indirect:

- Two types of routers: terminal and intermediate
- Terminal routers can source/sink traffic from nodes connected to them
- Intermediate routers propagate traffic between terminal routers

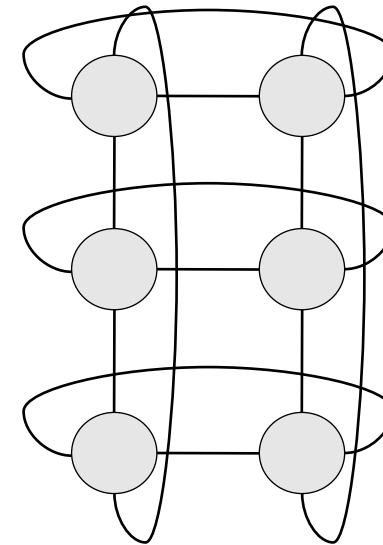
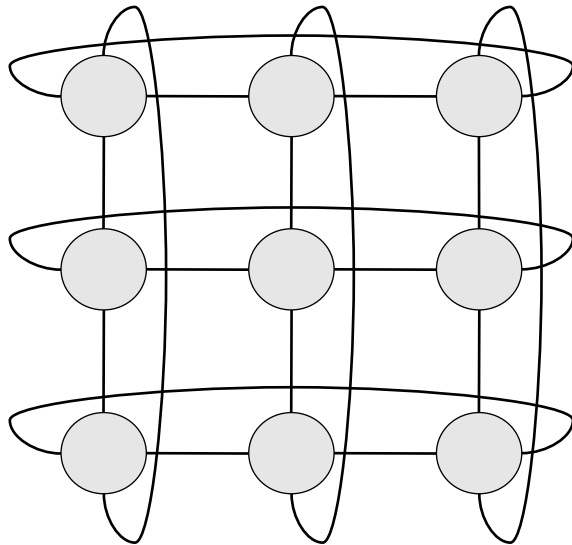
Direct:

- Every router is connected to a node yet can propagate passing traffic
- Every router is both terminal and intermediate
- Most NoCs use direct topologies

Direct Topologies – Torus

k-ary n-cube:

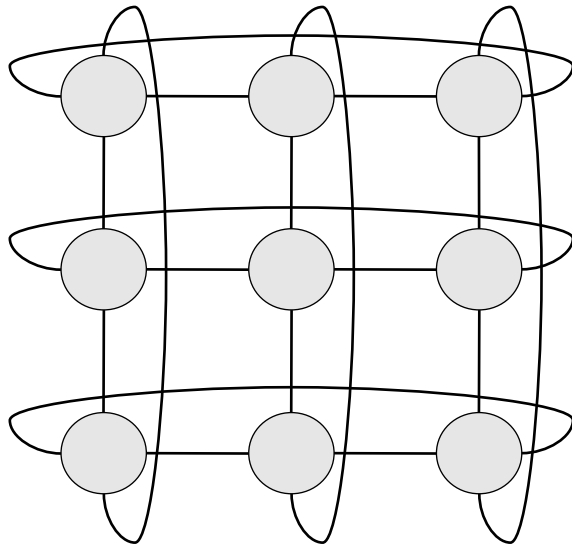
- n-dimensional grid with k routers in each dimension
- N routers = k^n (or $k_1 \times \dots \times k_n$ if not equilateral)



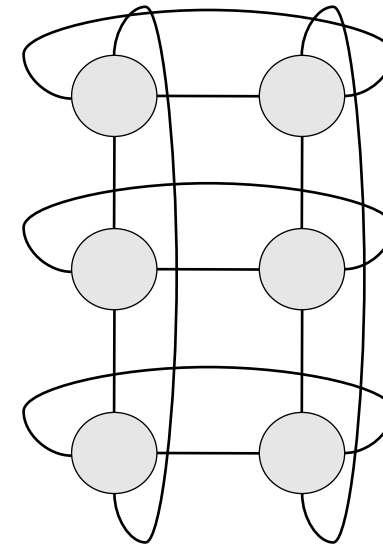
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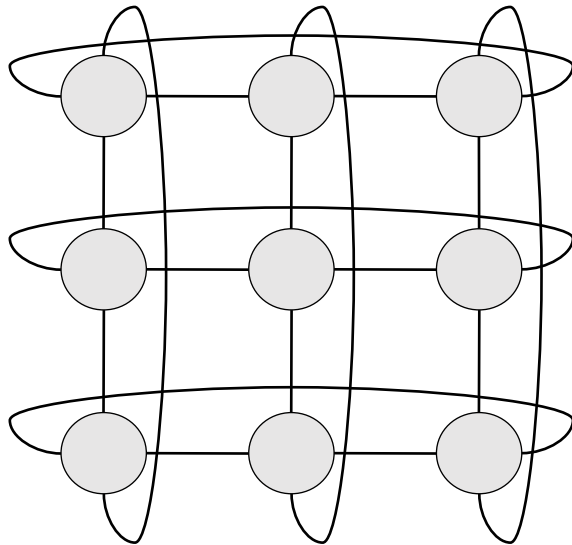
3-ary 2-cube



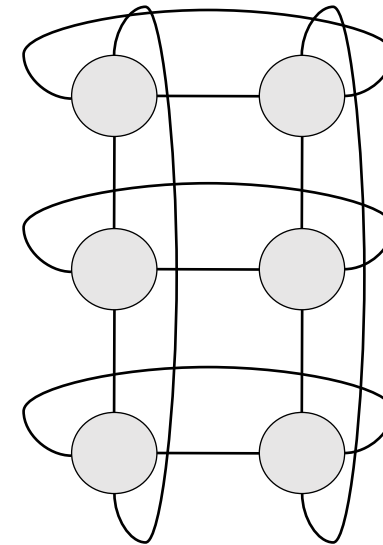
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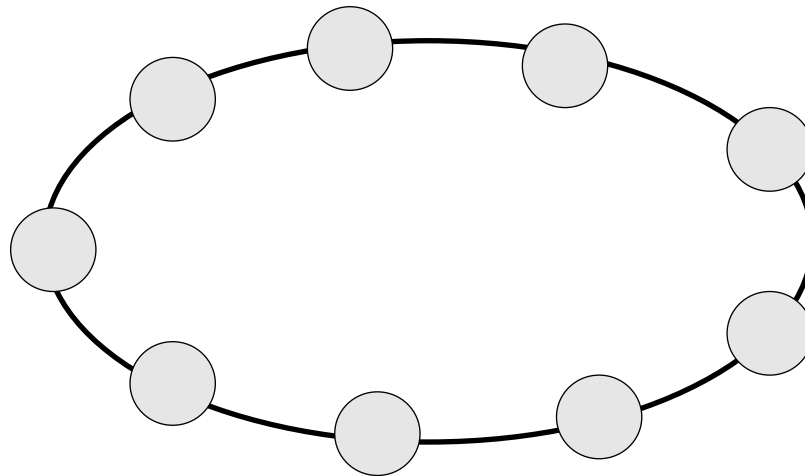
2,3-ary 2-cube



Direct Topologies – Torus

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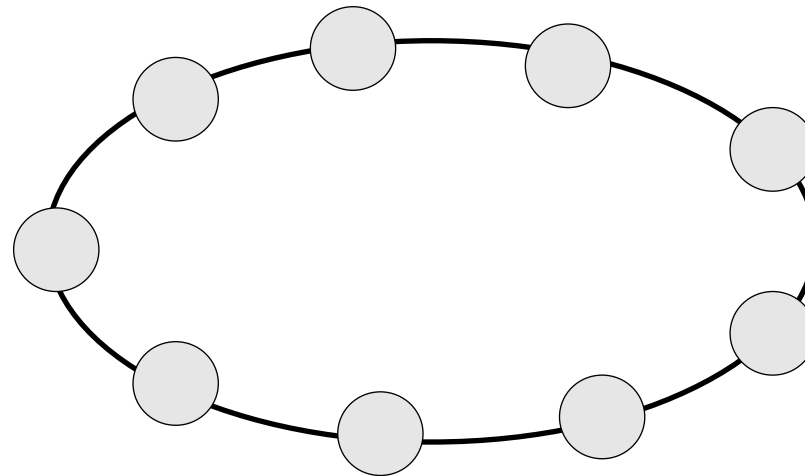
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Direct Topologies – Torus

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9-ary 1-cube
(i.e., ring)

Direct Topologies – Torus

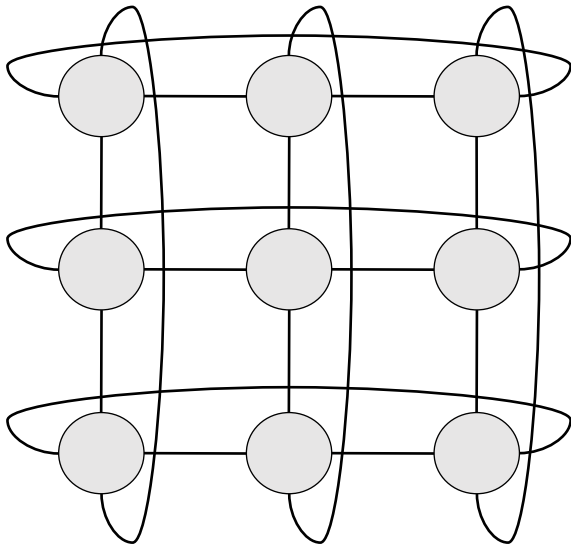
Advantages:

- Good for load balancing since edge-symmetric (due to wrap-around links)
- Good path diversity
- 1-dimensional and 2-dimensional map well to planar chip substrate

Direct Topologies – Torus

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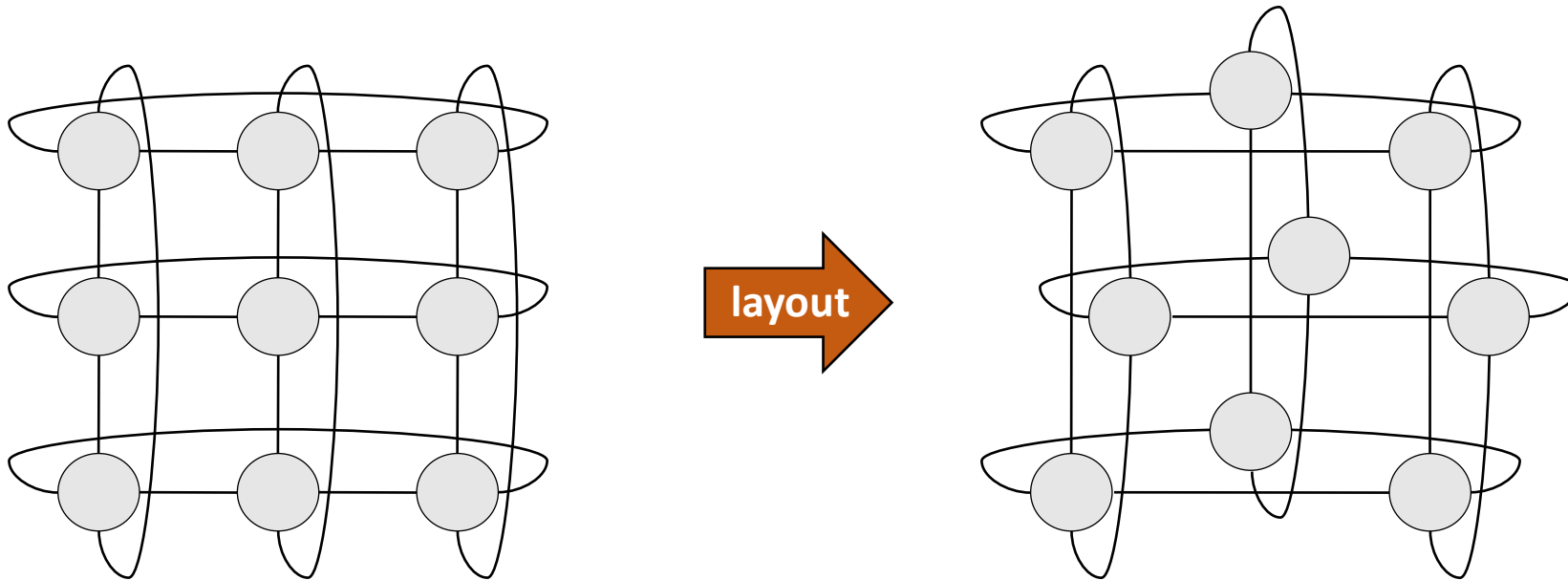
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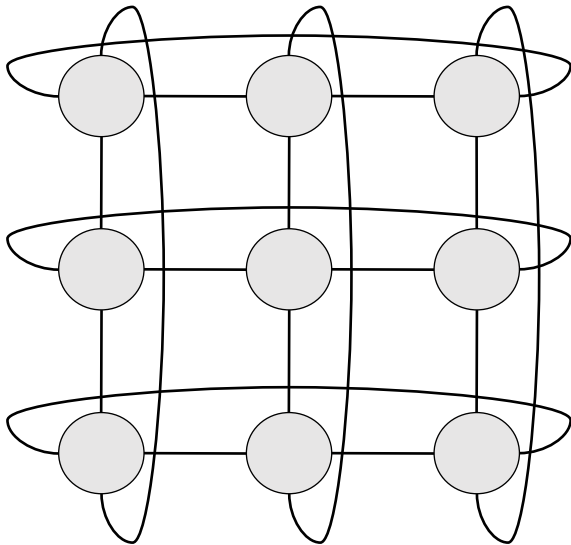
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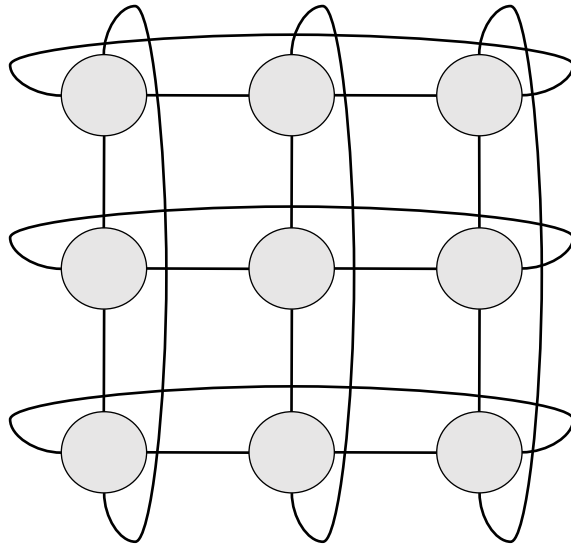
Degree:

- Degree = $2n$ bidirectional links, for all routers
- Total unidirectional links in network = $2nN$



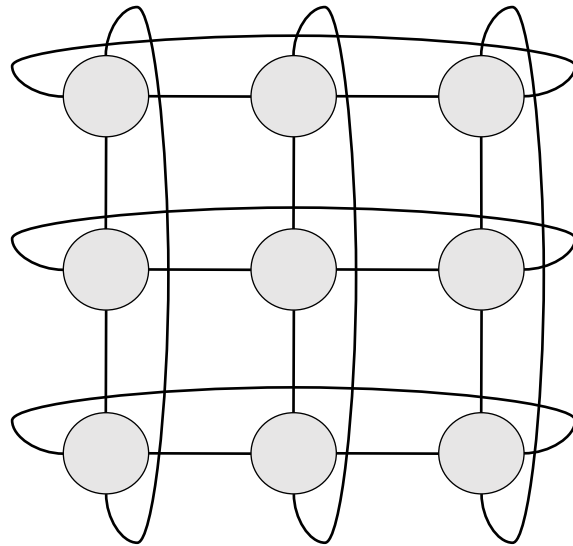
Direct Topologies – Torus

Network Diameter:



Direct Topologies – Torus

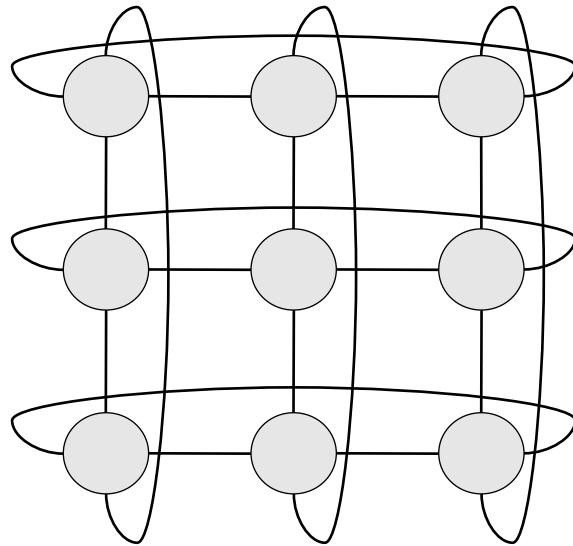
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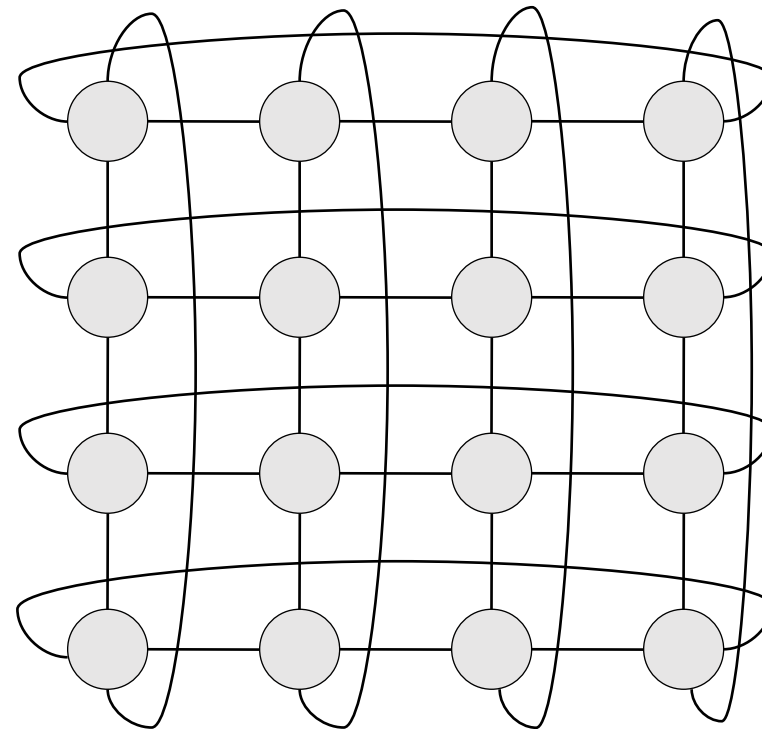
2 hops

Direct Topologies – Torus

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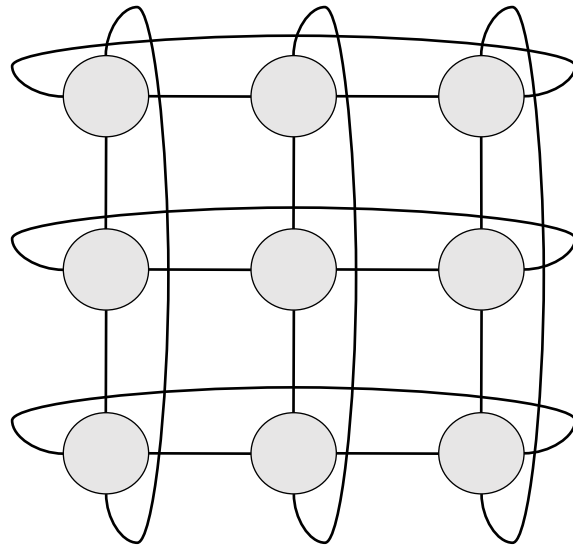
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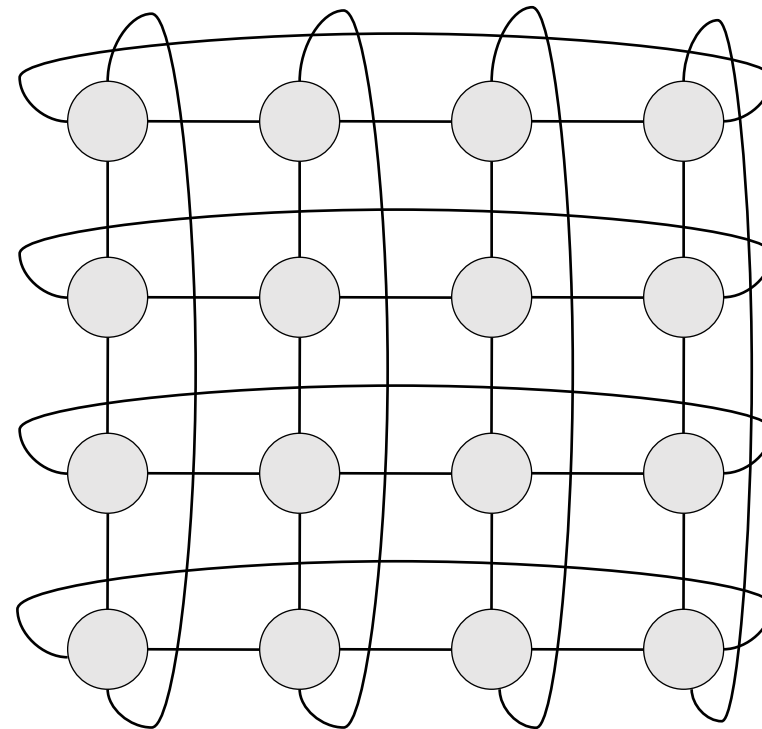
Direct Topologies – Torus

Network Diameter:

$$\text{diameter} = \begin{cases} n \left(\frac{k}{2} \right) & k \text{ even} \\ n \left(\frac{k-1}{2} \right) & k \text{ odd} \end{cases}$$



2 hops



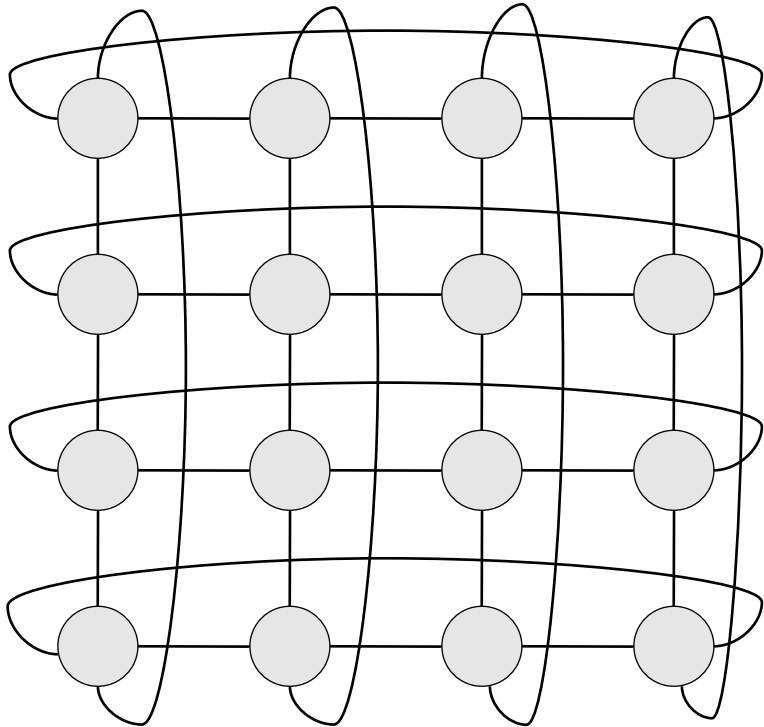
4 hops



Direct Topologies – Torus

Bisection Bandwidth:

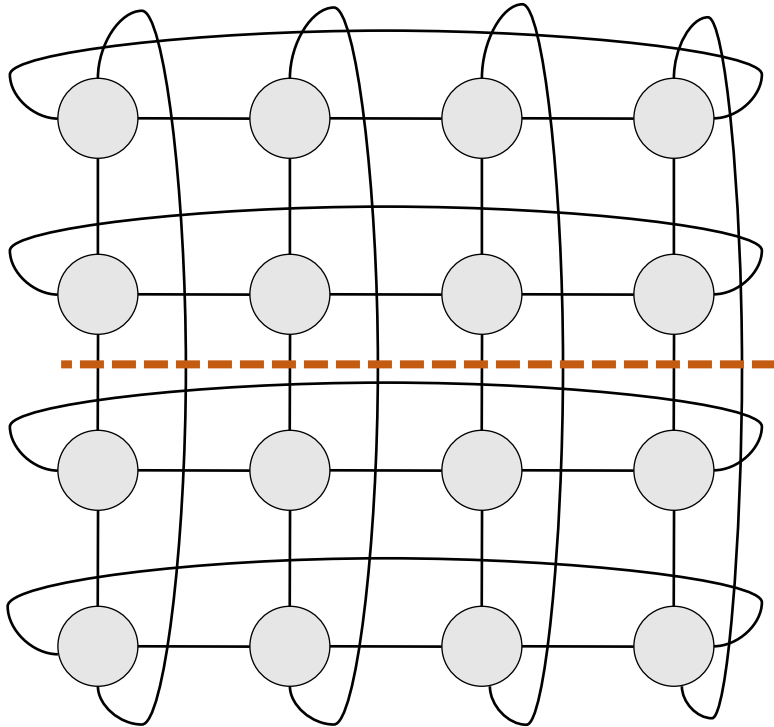
- Bisection bandwidth = $4N/k$ unidirectional links



Direct Topologies – Torus

Bisection Bandwidth:

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Two sets of N/k bidirectional links cross the bisection.

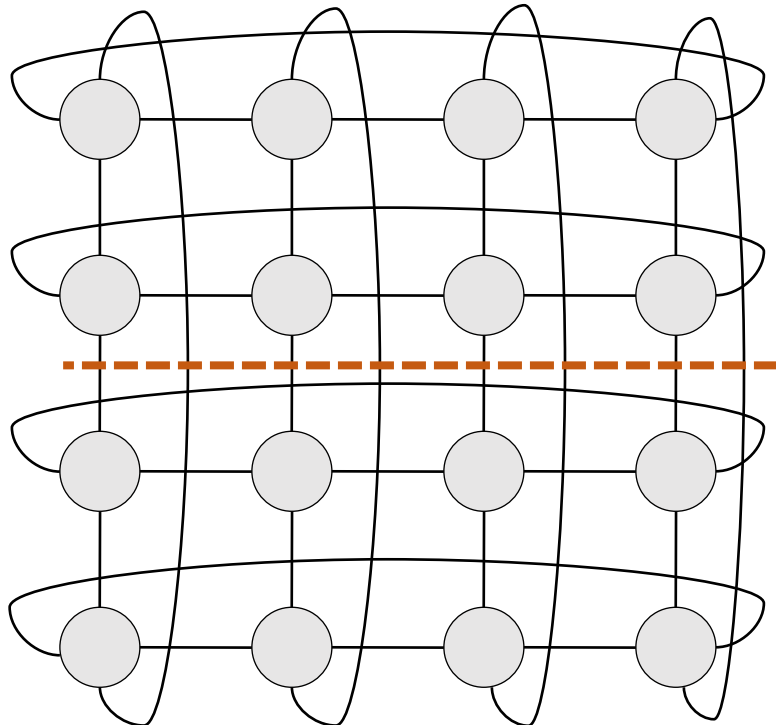
neighbor + wrap-around



Direct Topologies – Torus

Bisection Bandwidth:

- Bisection bandwidth = $4N/k$ unidirectional links



16 unidirectional links

Two sets of N/k bidirectional links cross the bisection.

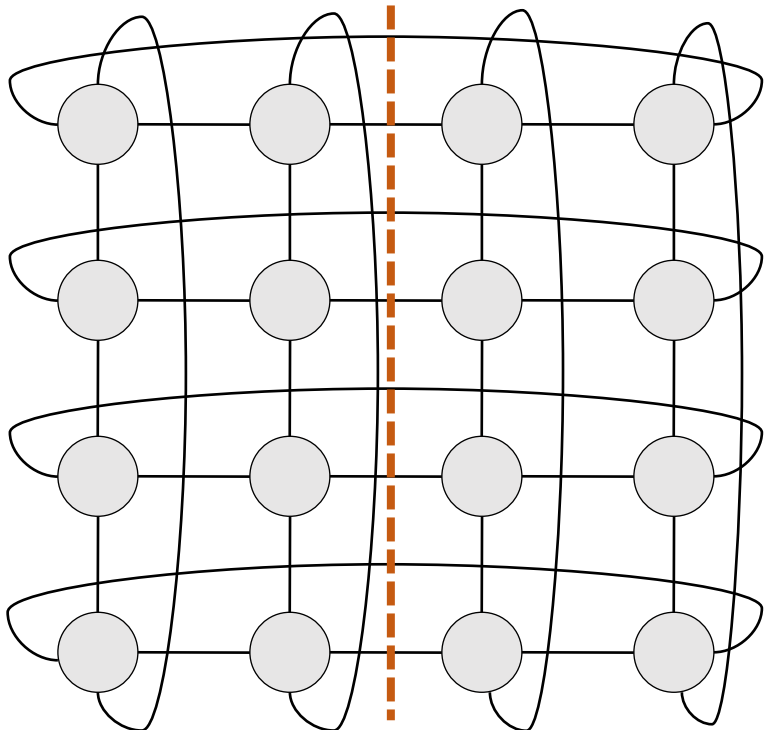
neighbor + wrap-around



Direct Topologies – Torus

Bisection Bandwidth:

- Bisection bandwidth = $4N/k$ unidirectional links



16 unidirectional links

Two sets of N/k bidirectional links cross the bisection.

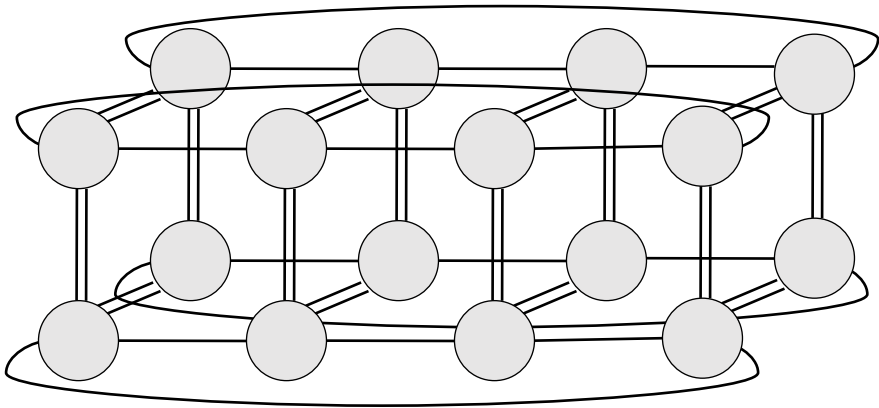
neighbor + wrap-around



Direct Topologies – Torus

Bisection Bandwidth:

- Bisection bandwidth = $4N/k$ unidirectional links



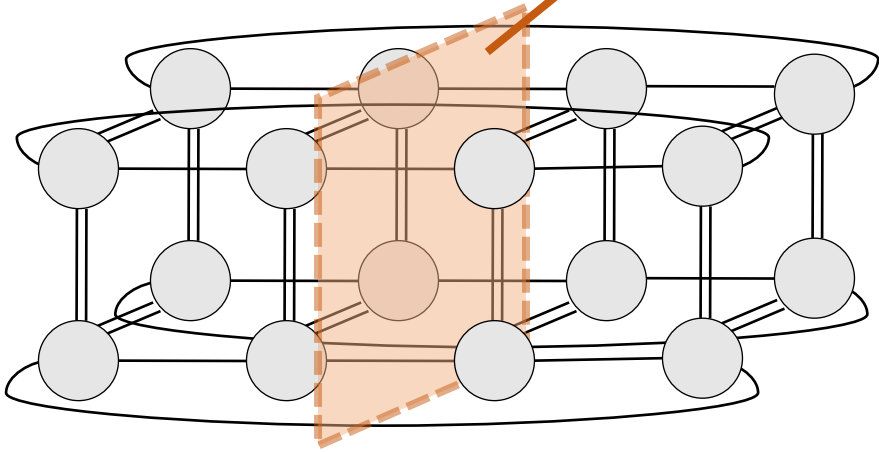
Direct Topologies – Torus

Bisection Bandwidth:

- Bisection bandwidth = $4N/k$ unidirectional links, **largest k**

$$N = k_1 k_2 \dots k_n$$

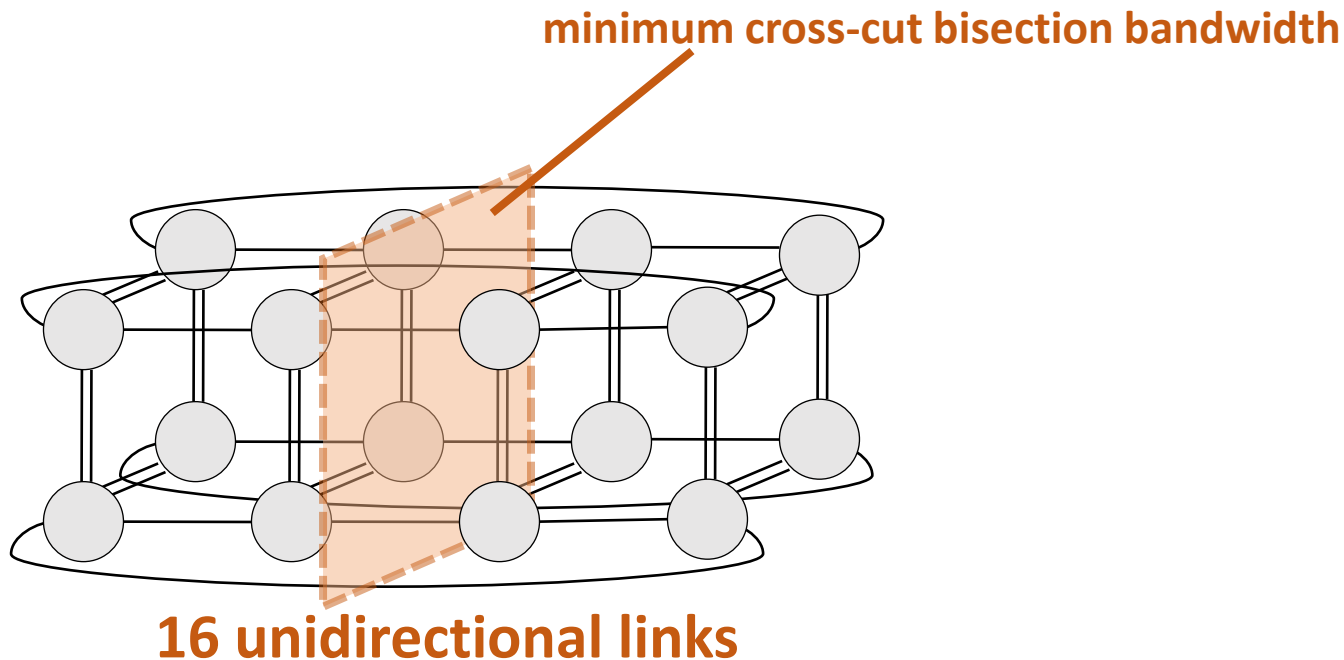
minimum cross-cut bisection bandwidth



Direct Topologies – Torus

Bisection Bandwidth:

- Bisection bandwidth = $4N/k$ unidirectional links, **largest k**
- $N = k_1 k_2 \dots k_n$ 4



Direct Topologies – Torus

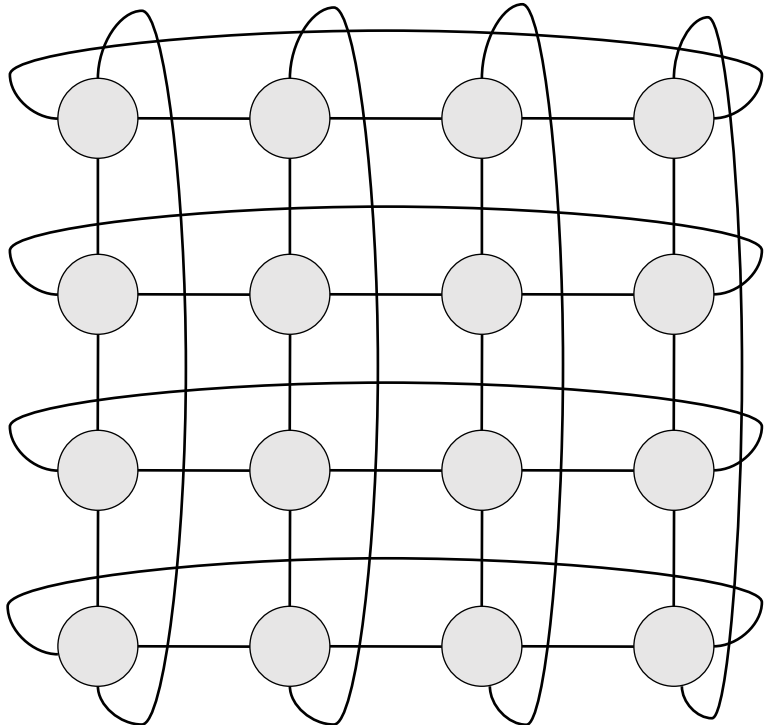
Bisection Traffic:

- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

Direct Topologies – Torus

Bisection Traffic:

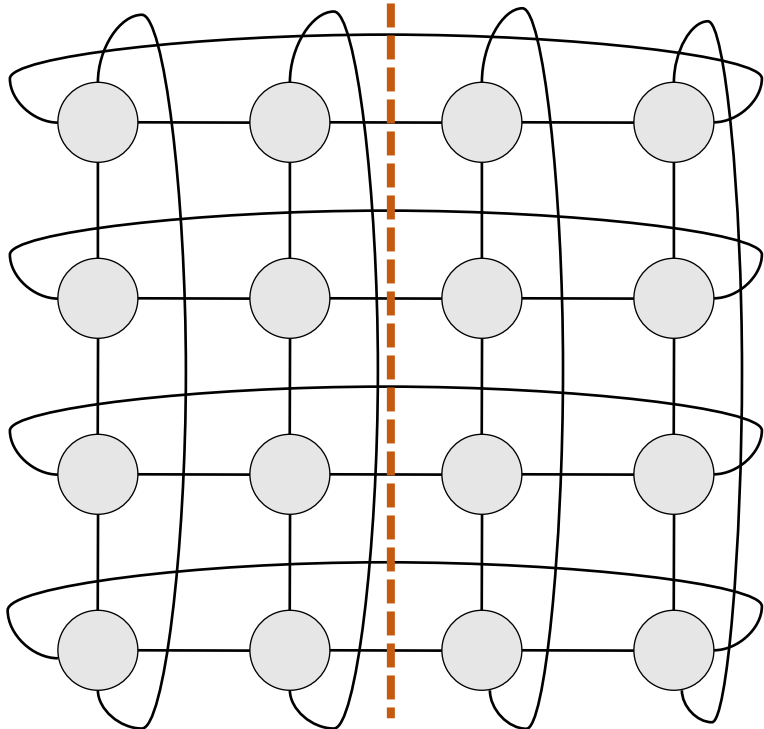
- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



Direct Topologies – Torus

Bisection Traffic:

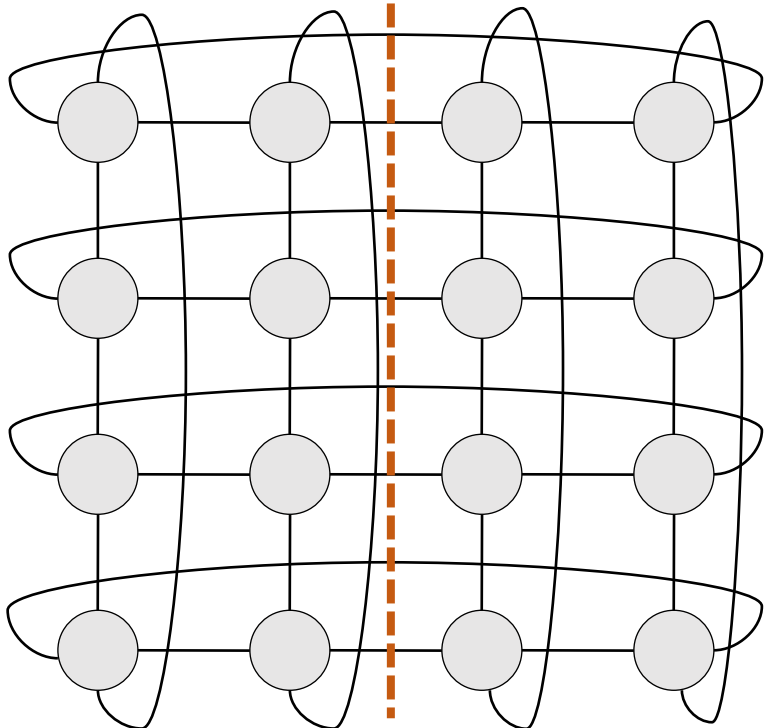
- Assume even number of routers N
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- Assume injection rate of 1 packet/router/cycle



Direct Topologies – Torus

Bisection Traffic:

- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



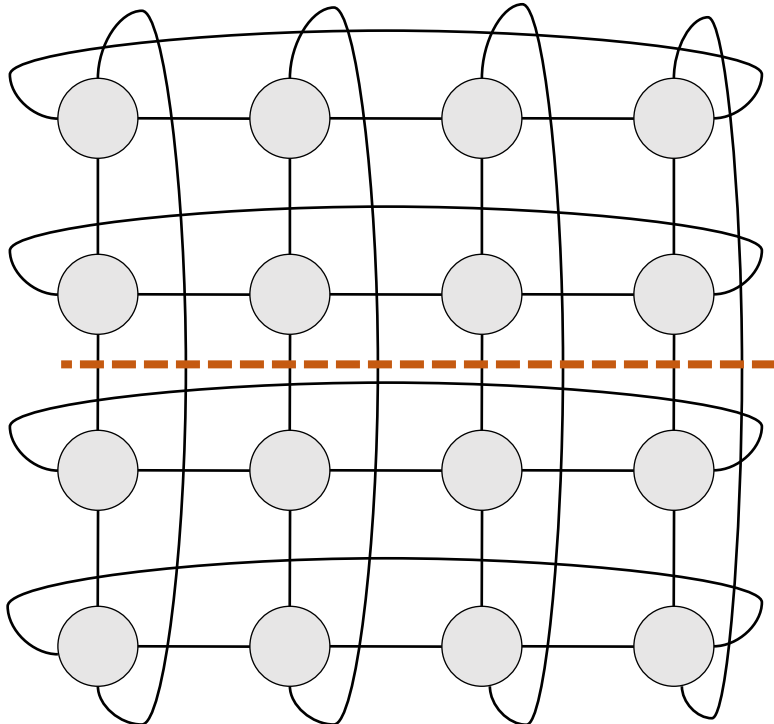
$N/2 = 8$ packets/cycle



Direct Topologies – Torus

Bisection Traffic:

- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



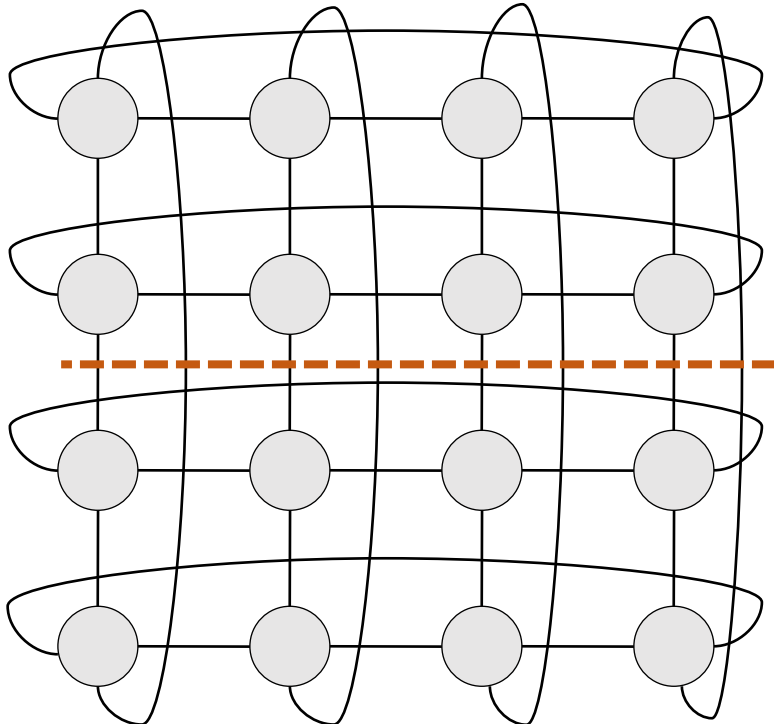
$N/2 = 8$ packets/cycle



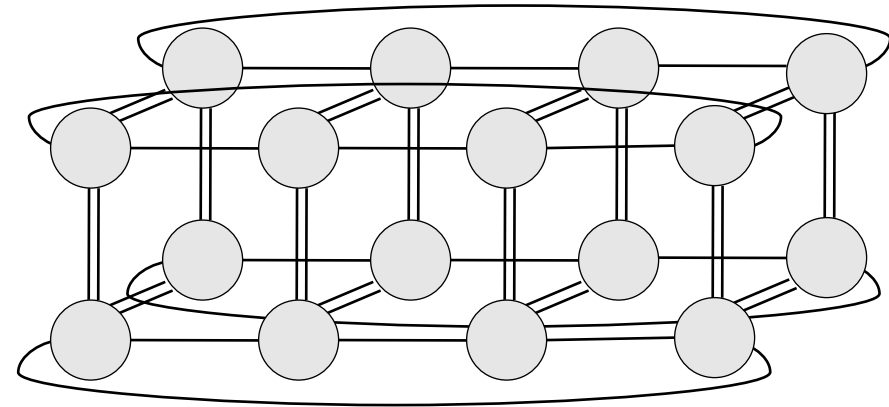
Direct Topologies – Torus

Bisection Traffic:

- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



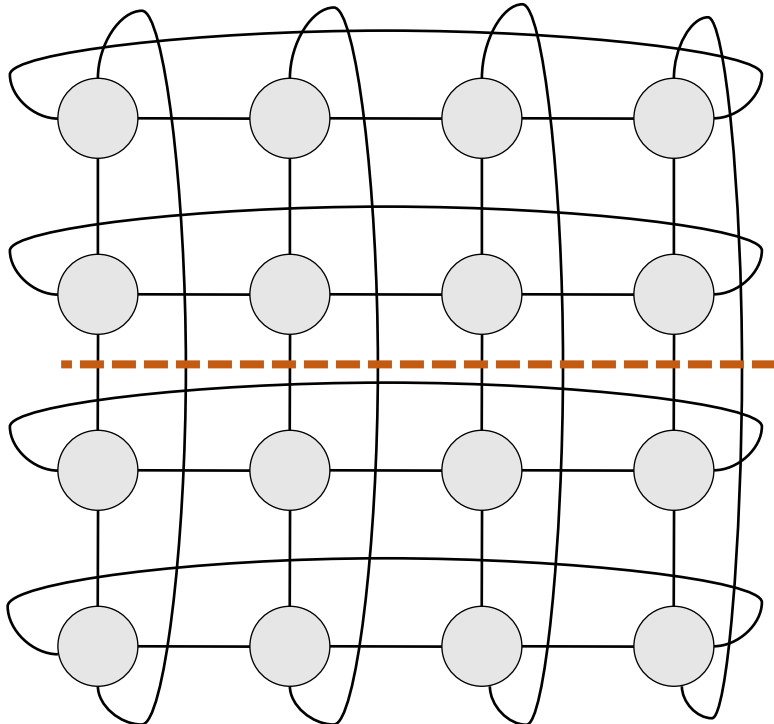
$N/2 = 8$ packets/cycle



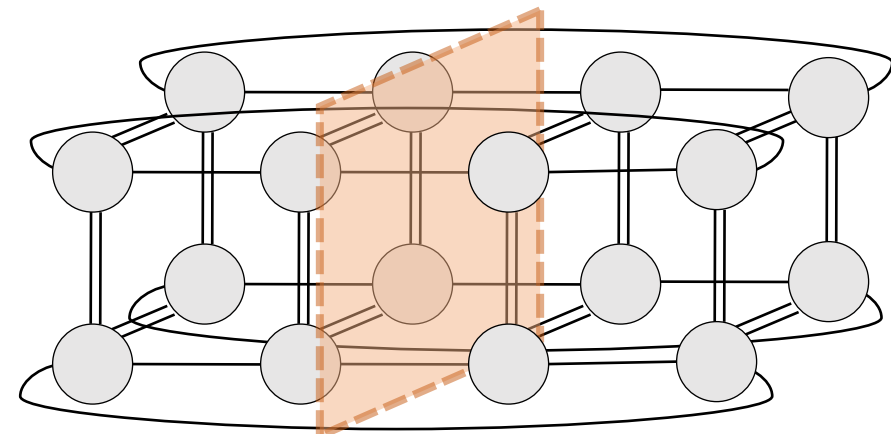
Direct Topologies – Torus

Bisection Traffic:

- Assume even number of routers N
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



$N/2 = 8$ packets/cycle



$N/2 = 8$ packets/cycle



Direct Topologies – Torus

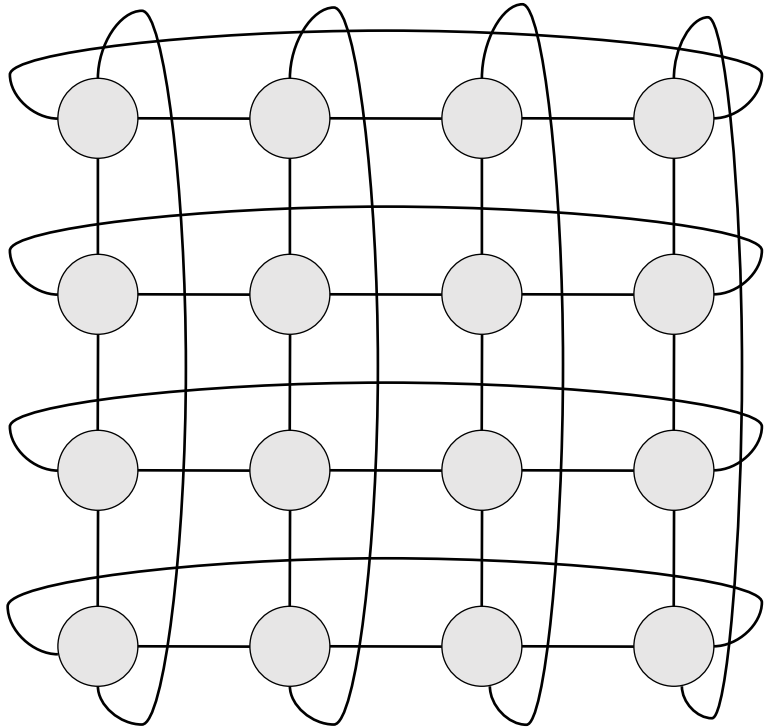
Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

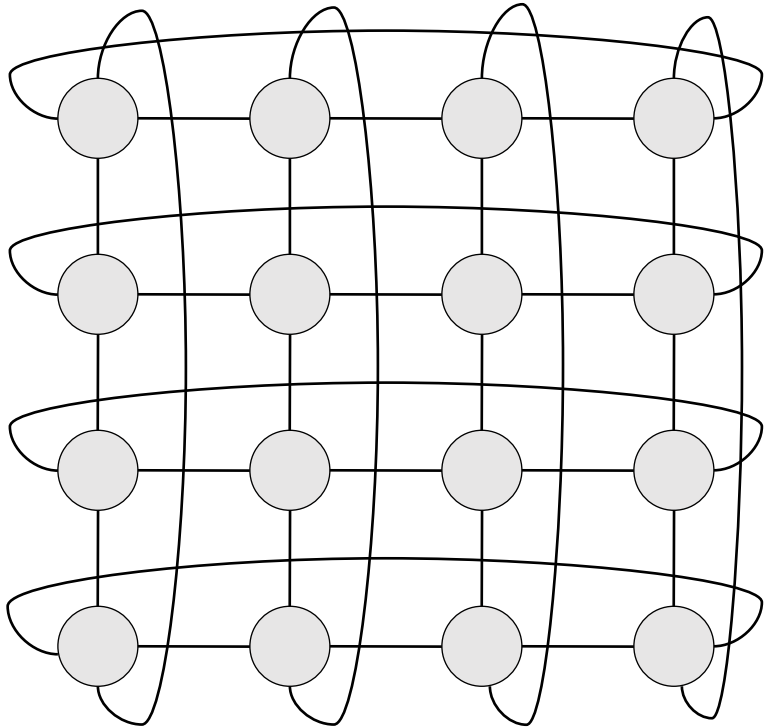


What's the maximum channel load?

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

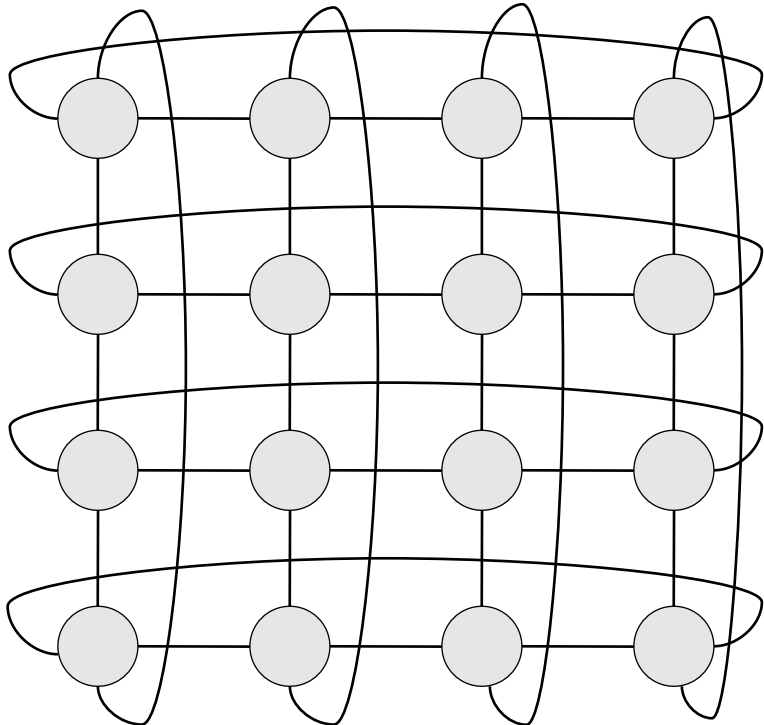


***What's the maximum channel load?
Which are the most congested links?***

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

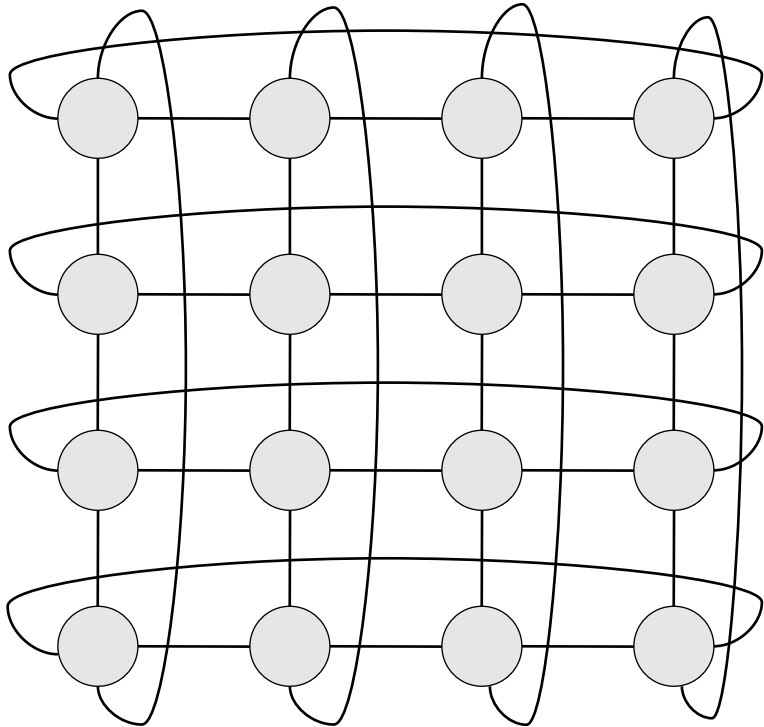


What's the maximum channel load?
Which are the most congested links?
All are equally congested

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

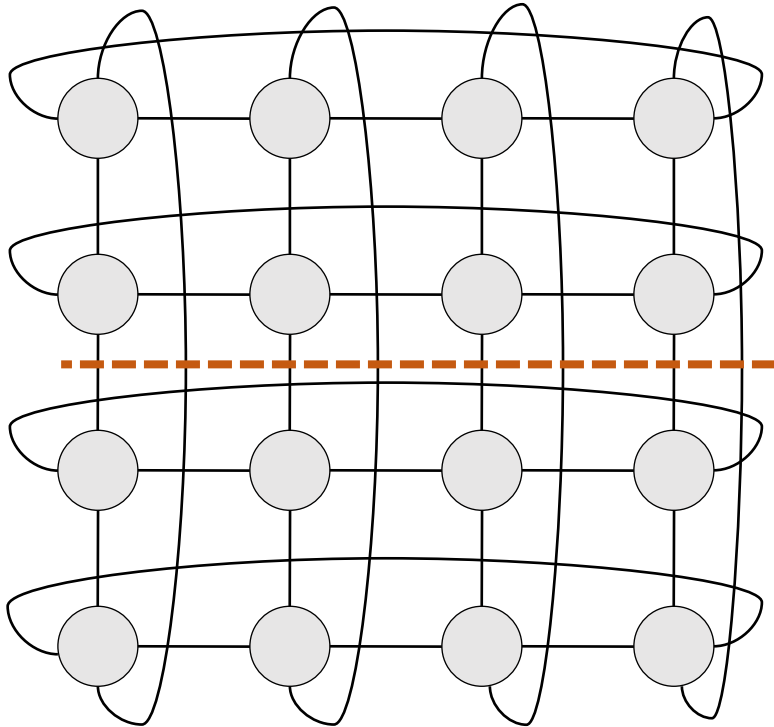
- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

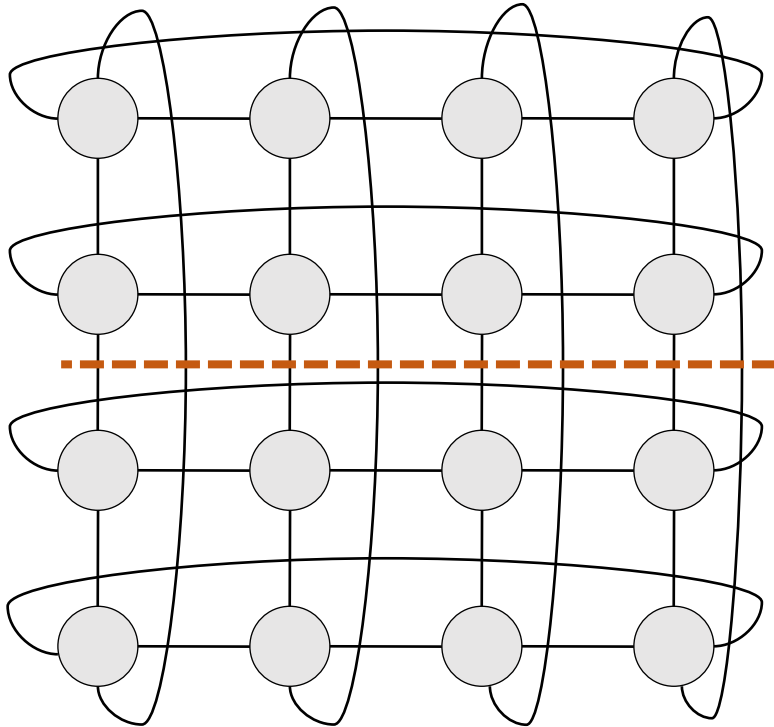
- Assume even number of routers per dimension k
- Assume uniform random traffic
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Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
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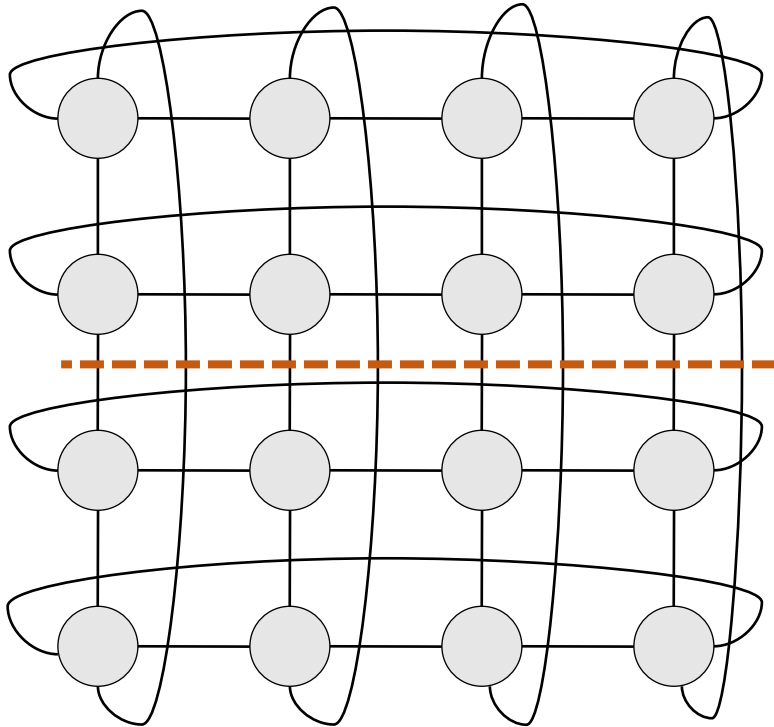


Channel load = bisection traffic / unidirectional links:

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



Channel load = bisection traffic / unidirectional links:

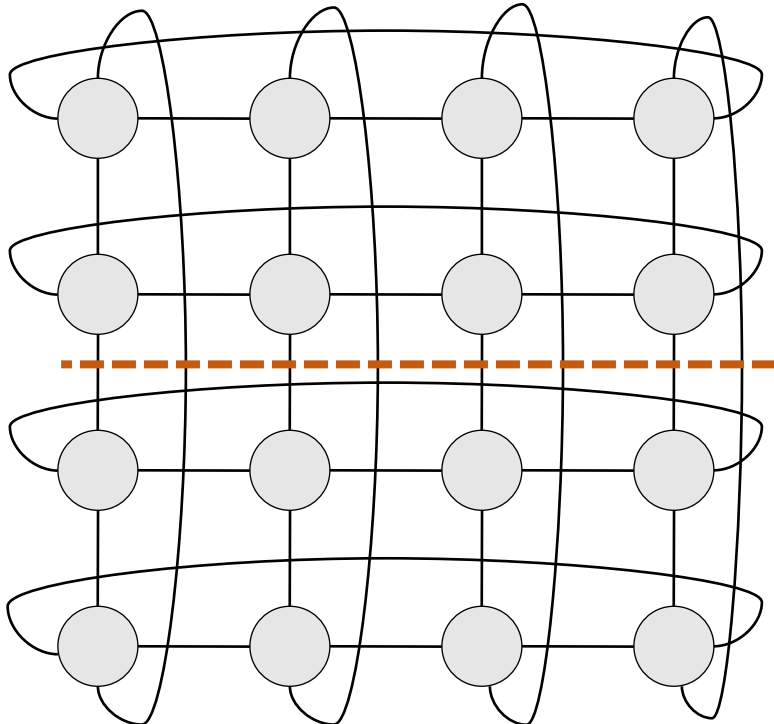
$$\text{channel load} = \frac{N}{2} \times \frac{k}{4N} = \frac{k}{8}$$



Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



$k/8 = 1$ packet per 2 cycles

Channel load = bisection traffic / unidirectional links:

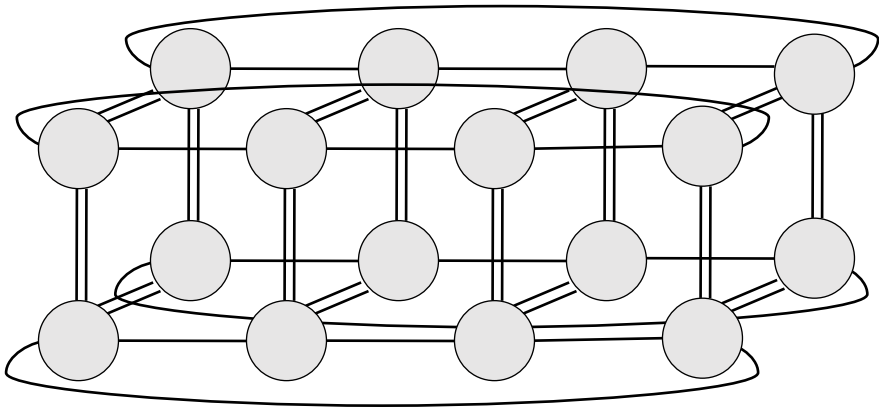
$$\text{channel load} = \frac{N}{2} \times \frac{k}{4N} = \frac{k}{8}$$



Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

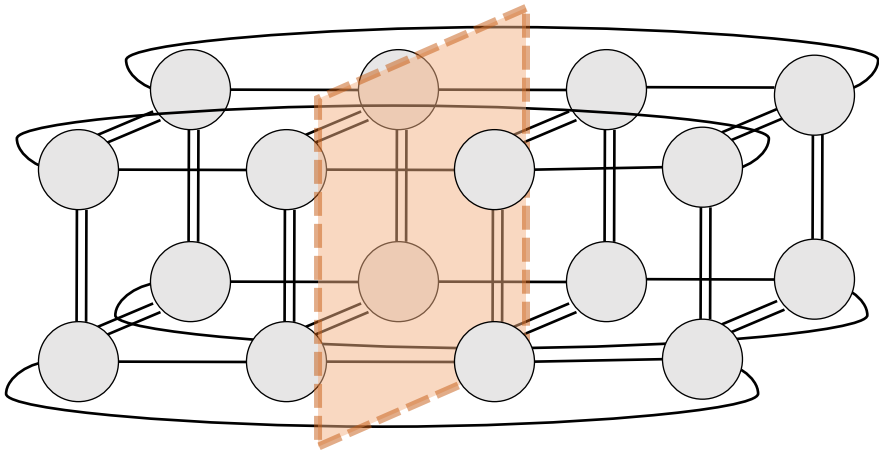


*What's the maximum channel load?
Which are the most congested links?*

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



What's the maximum channel load?

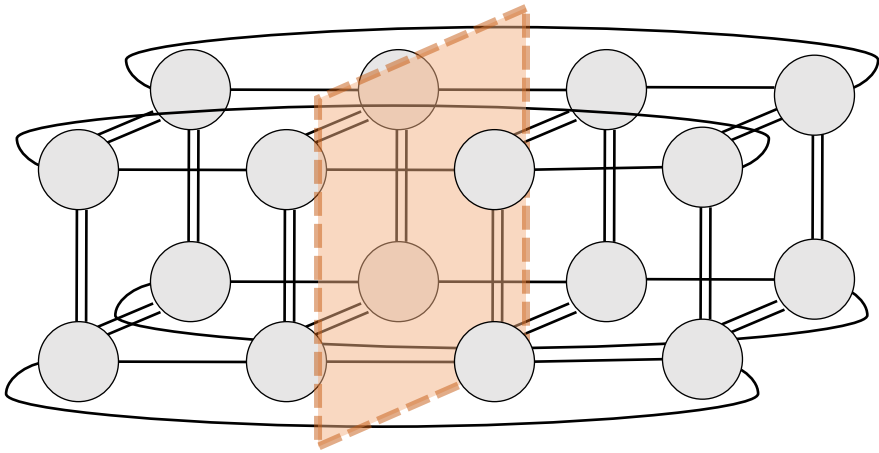
Which are the most congested links?

Minimum bisection (i.e., largest dimension $k=4$)

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle

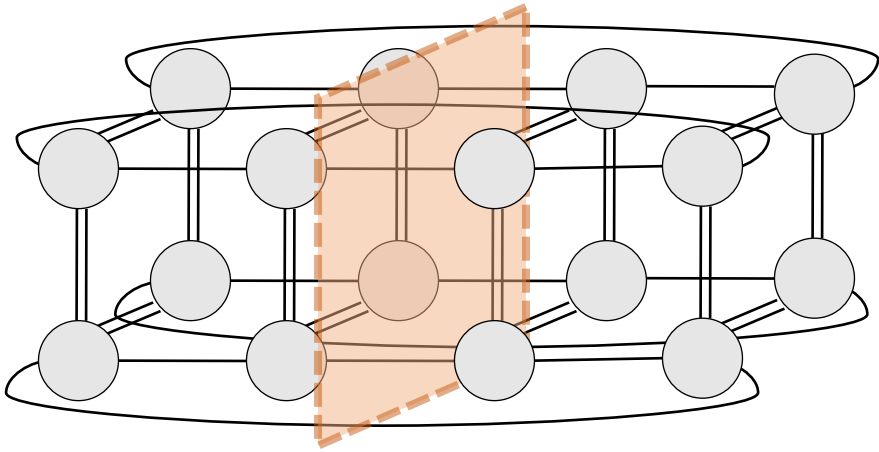


Channel load = bisection traffic / unidirectional links:

Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



Channel load = bisection traffic / unidirectional links:

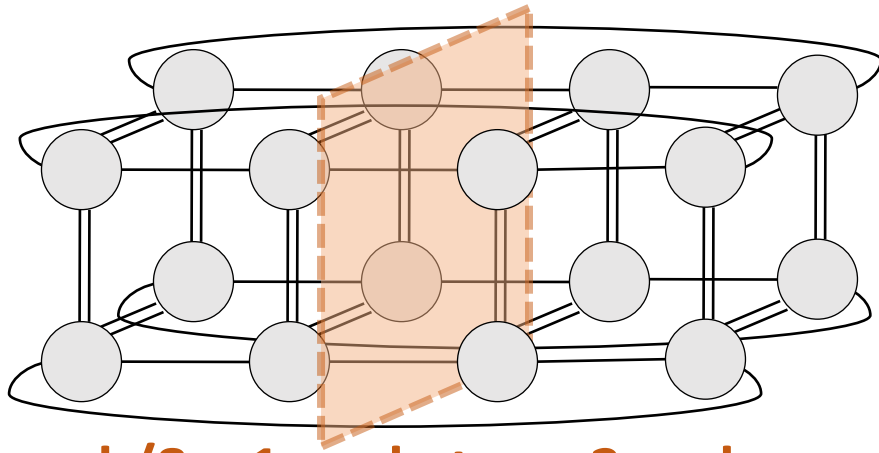
$$\text{channel load} = \frac{N}{2} \times \frac{k}{4N} = \frac{k}{8}$$



Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



$k/8 = 1$ packet per 2 cycles

Channel load = bisection traffic / unidirectional links:

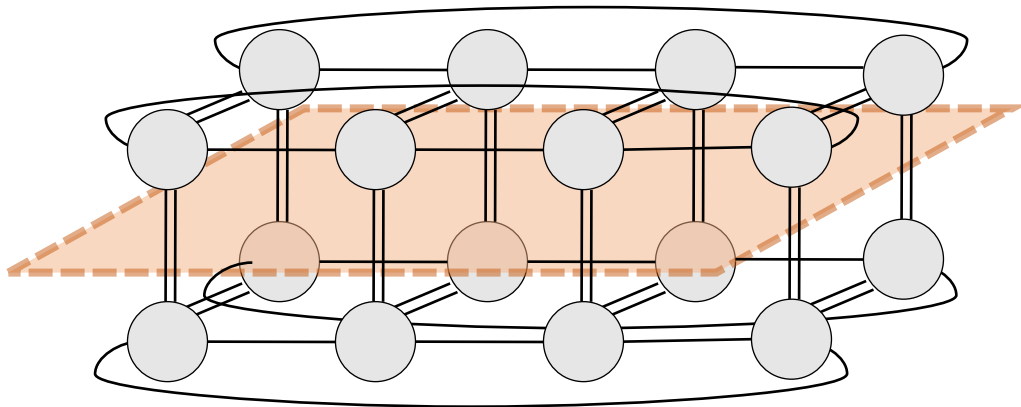
$$\text{channel load} = \frac{N}{2} \times \frac{k}{4N} = \frac{k}{8}$$



Direct Topologies – Torus

Maximum (i.e., Highest Average) Channel Load:

- Assume even number of routers per dimension k
- Assume uniform random traffic
- Assume injection rate of 1 packet/router/cycle



$k/8 = 1$ packet per 4 cycles

Channel load = bisection traffic / unidirectional links:

$$\text{channel load} = \frac{N}{2} \times \frac{k}{4N} = \frac{k}{8}$$



Direct Topologies – Torus

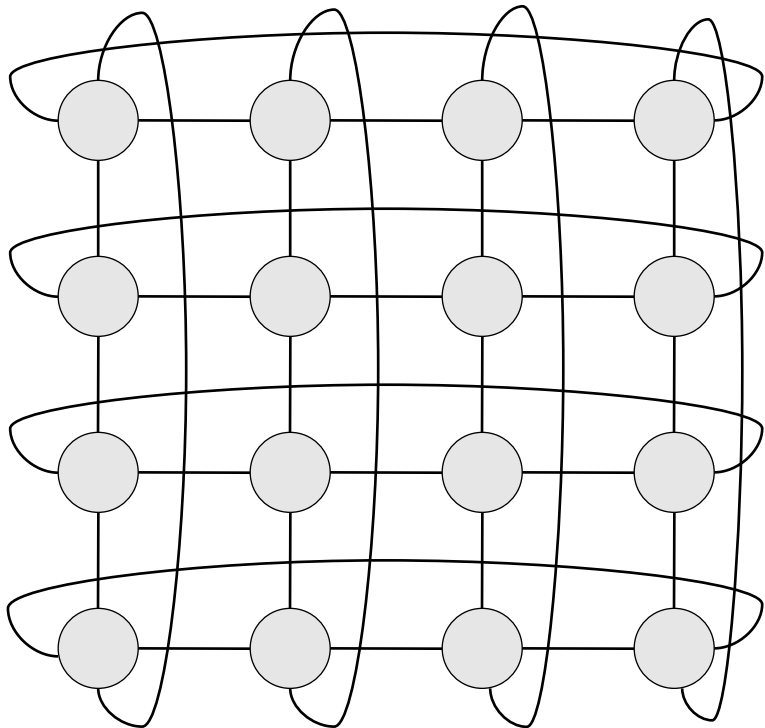
Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

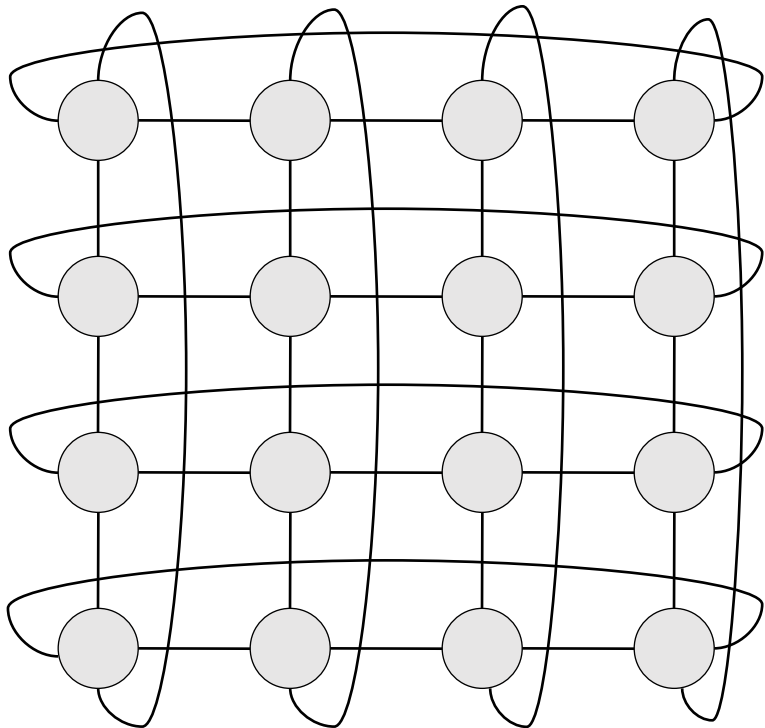


What's the maximum path diversity?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

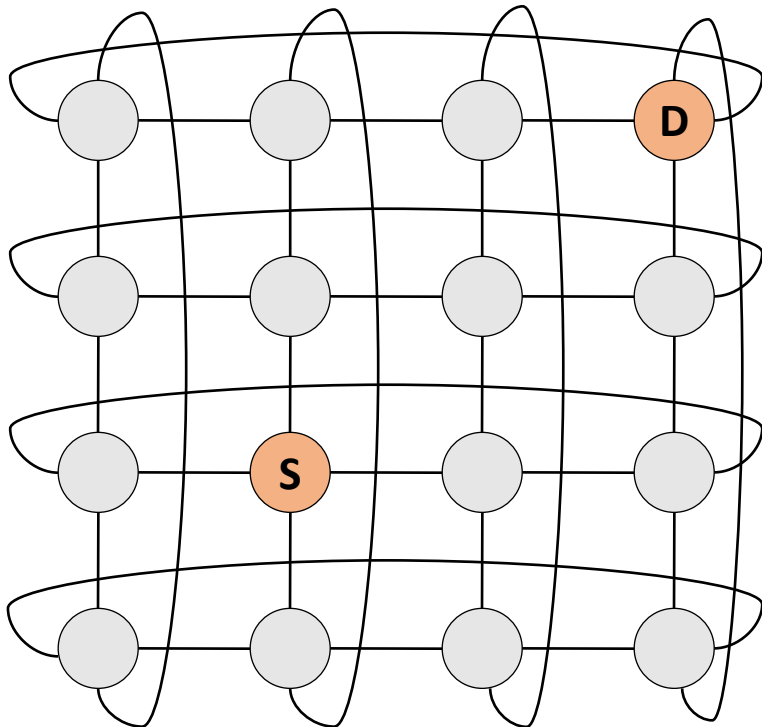


How many hops away is the furthest destination?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

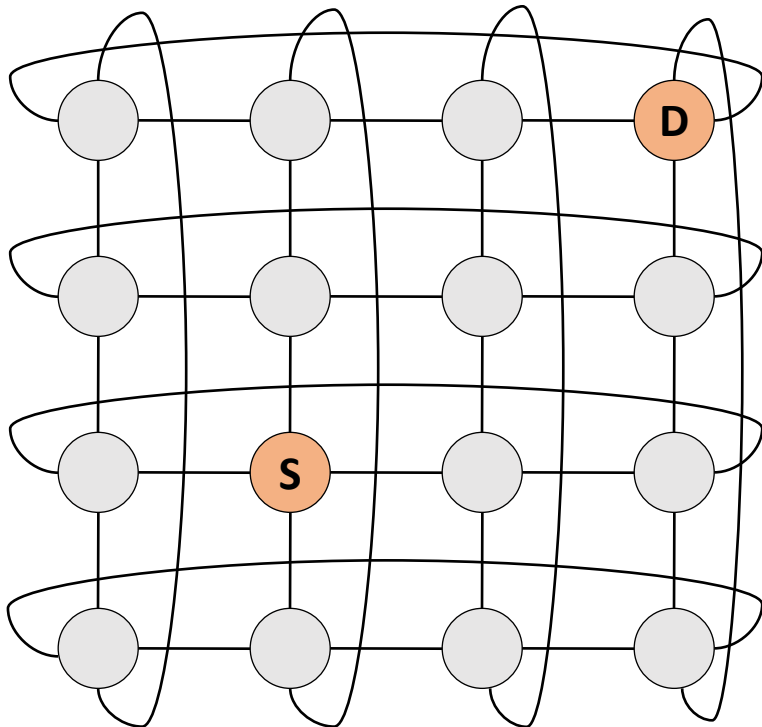


How many hops away is the furthest destination?
 $nk/2 = 4$ hops (i.e., network diameter)

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

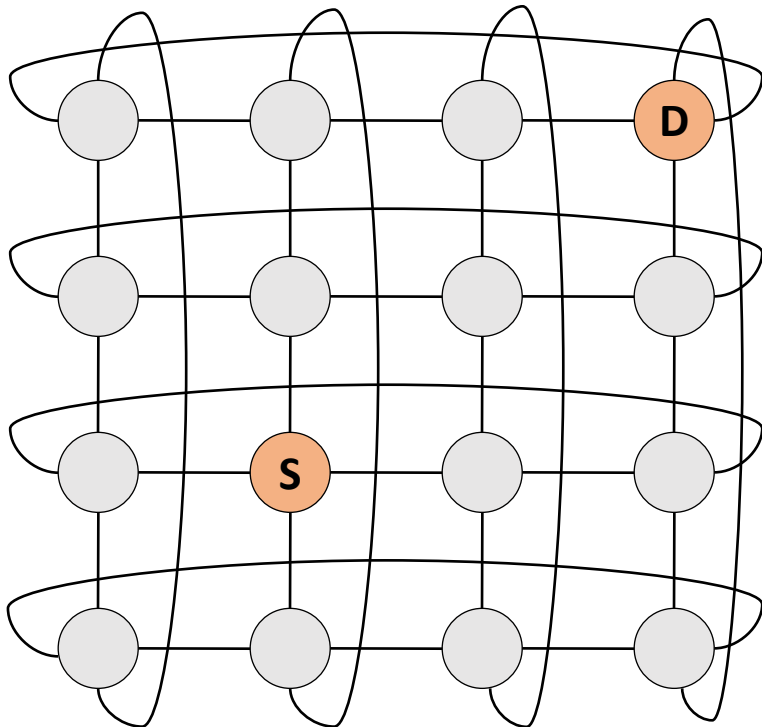


How many directions can we take?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



How many directions can we take?

$2^n = 4$ directions

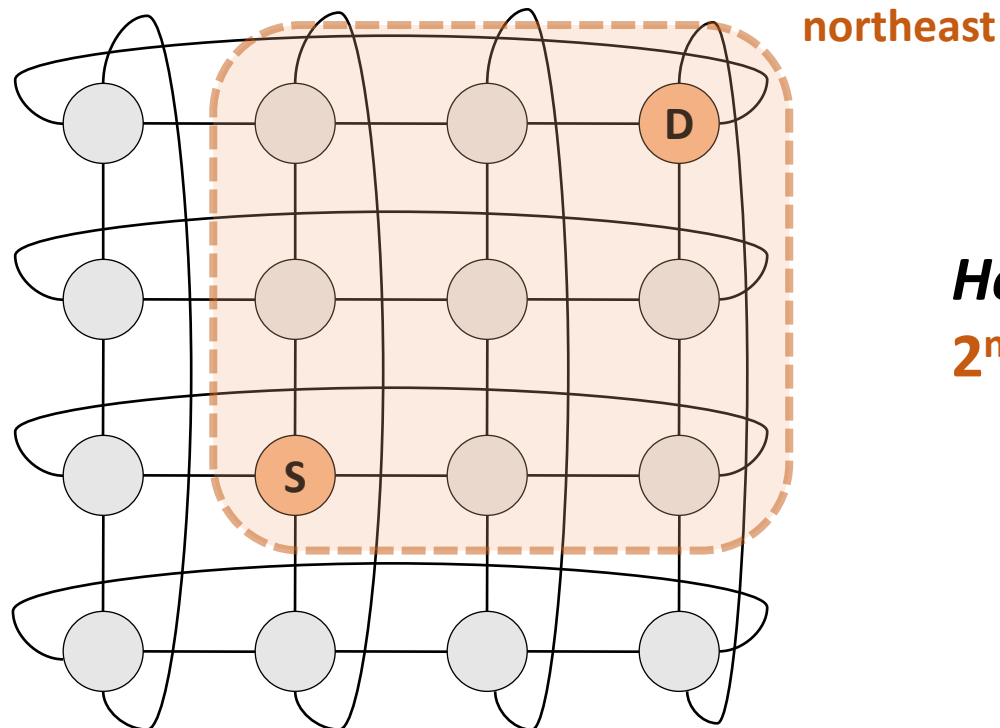
since edge-symmetric with even k,
all directions have same hop count



Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



How many directions can we take?

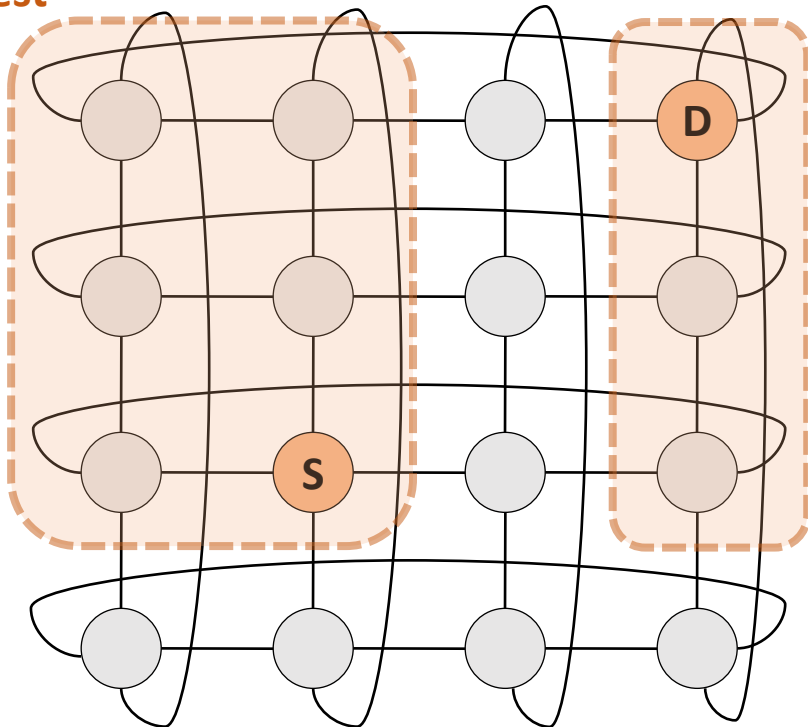
$2^n = 4$ directions

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

northwest



How many directions can we take?

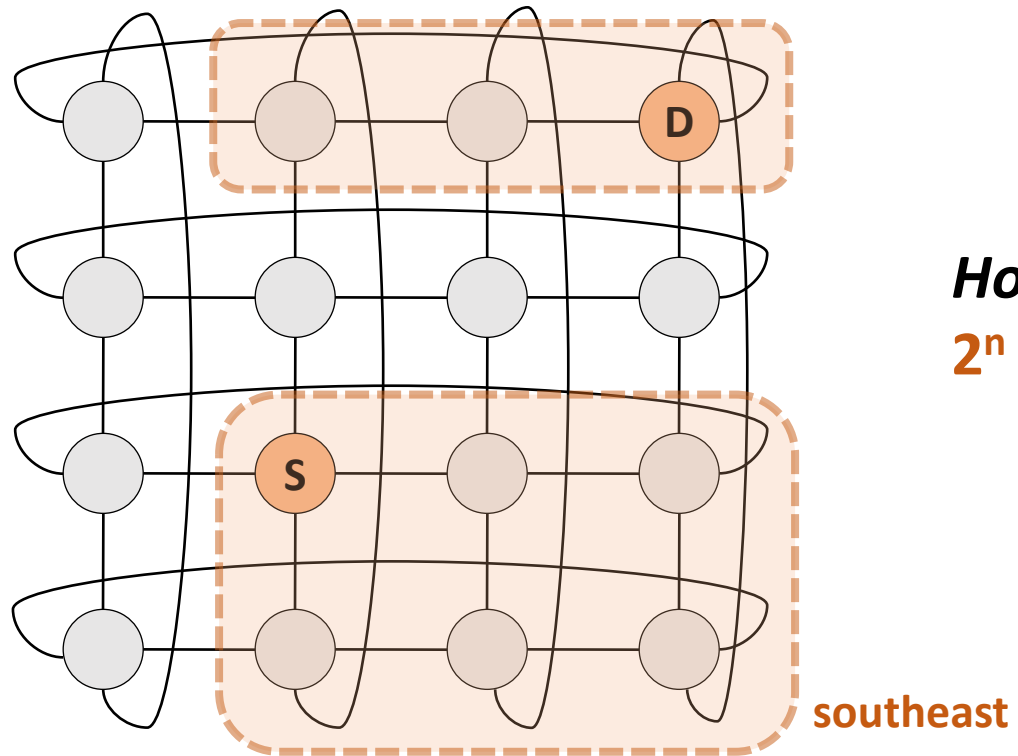
$2^n = 4$ directions



Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



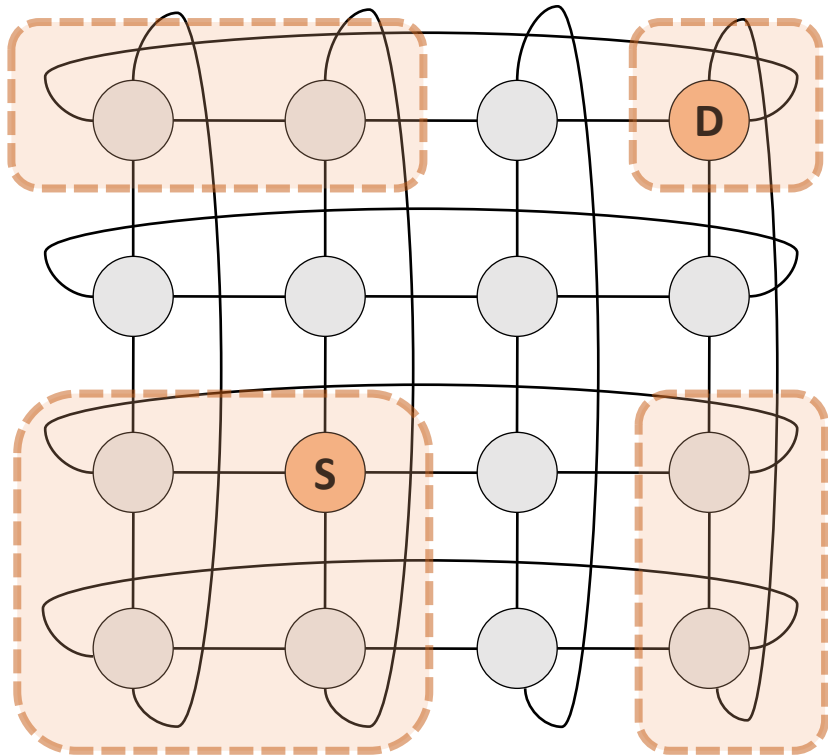
How many directions can we take?

$2^n = 4$ directions

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



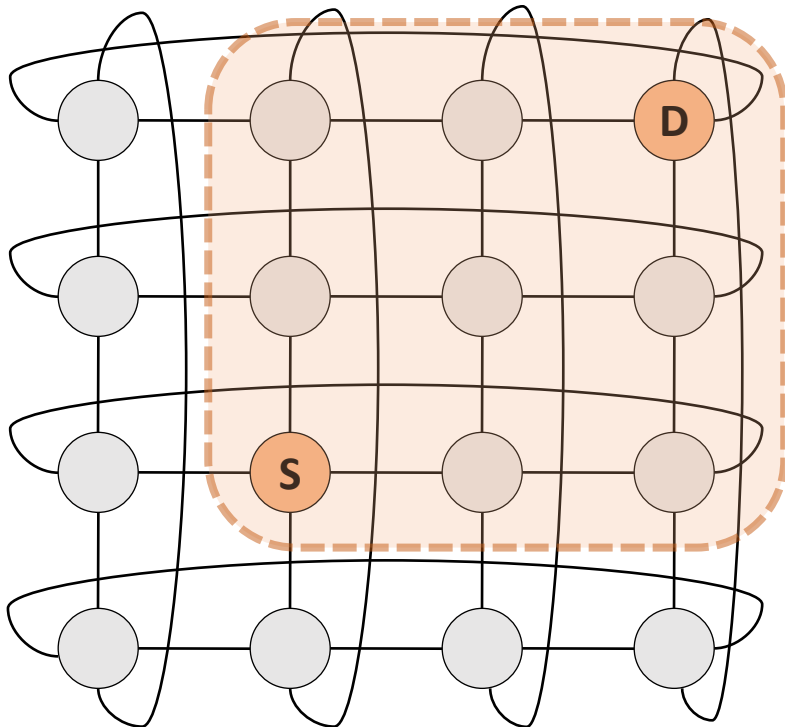
How many directions can we take?

$2^n = 4$ directions

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

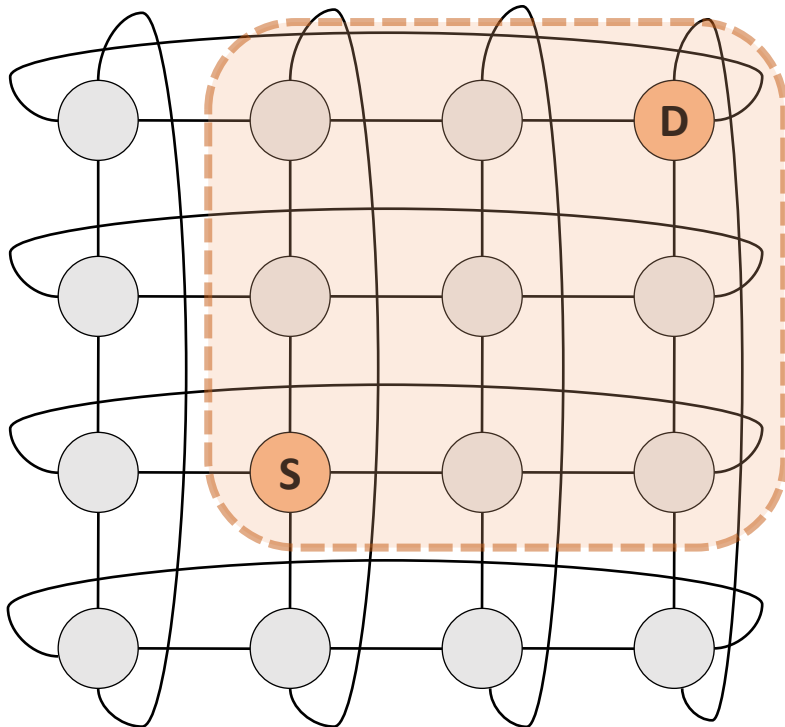


How many paths per direction do we have?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



How many paths per direction do we have?

6 paths per direction

$$\binom{n}{\frac{k}{2}} = 6$$

total hops

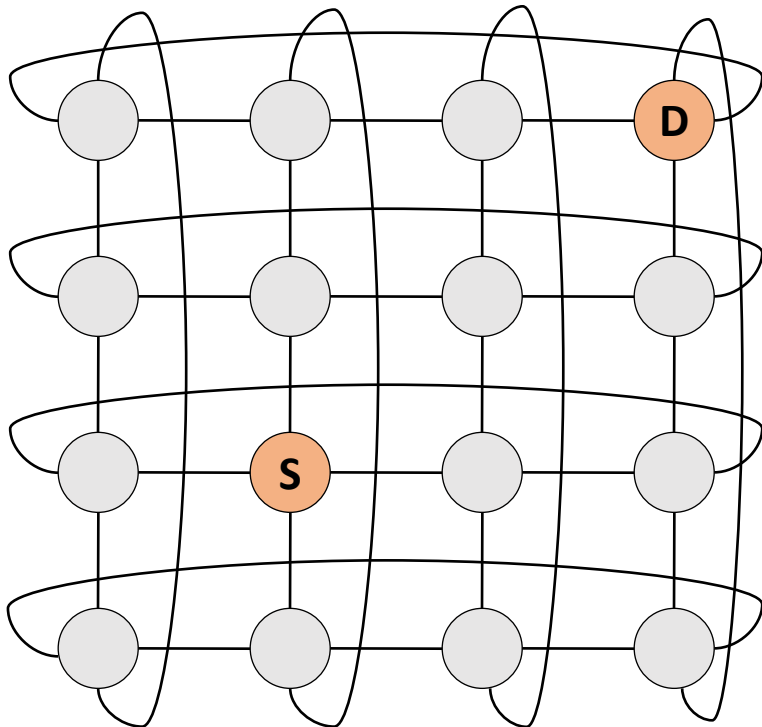
hops per dimension



Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

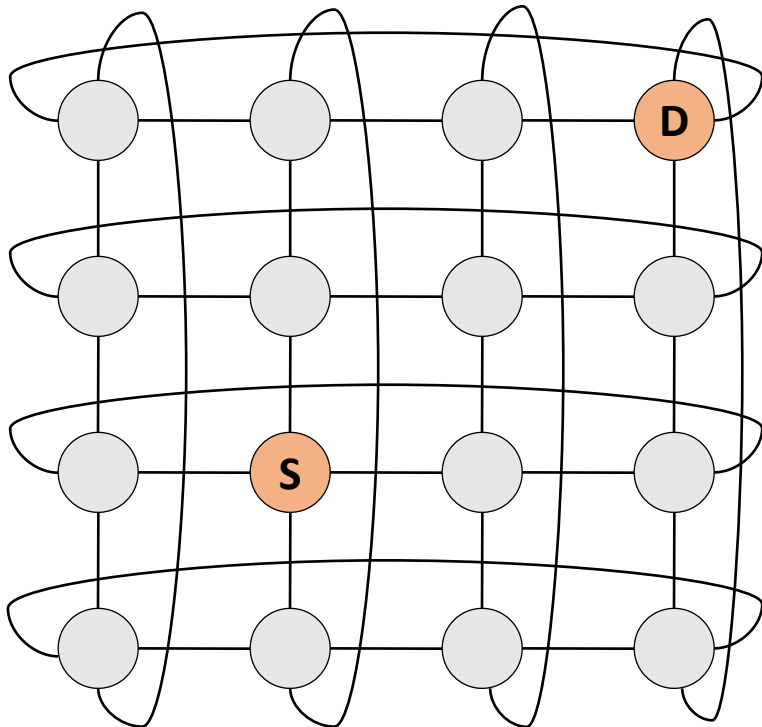


What's the maximum path diversity?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



What's the maximum path diversity?

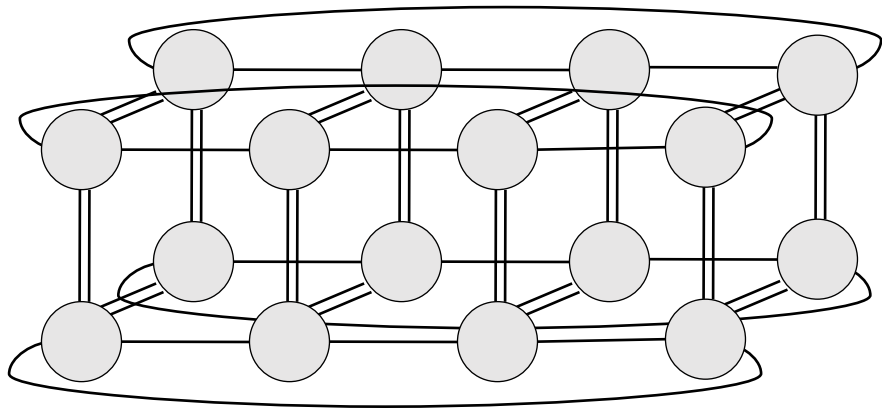
24 paths

$$\text{max path diversity} = 2^n \times \binom{n \left(\frac{k}{2} \right)}{\frac{k}{2}} = 4 \text{ directions} \times 6 \text{ paths} = 24$$

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

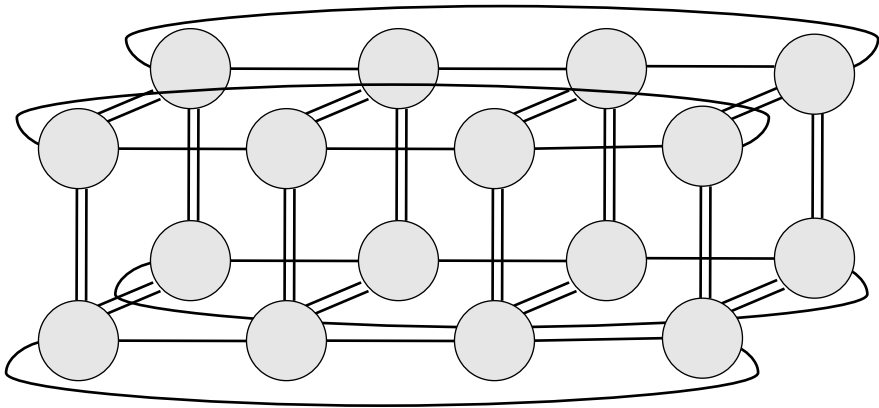


What's the maximum path diversity?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

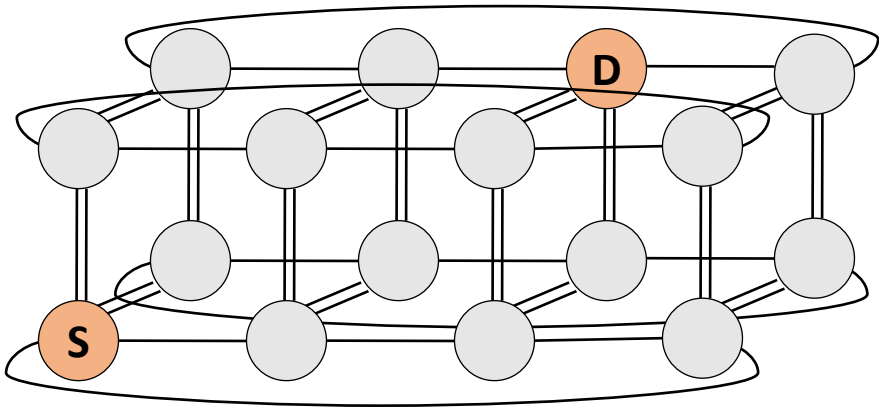


How many hops away is the furthest destination?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



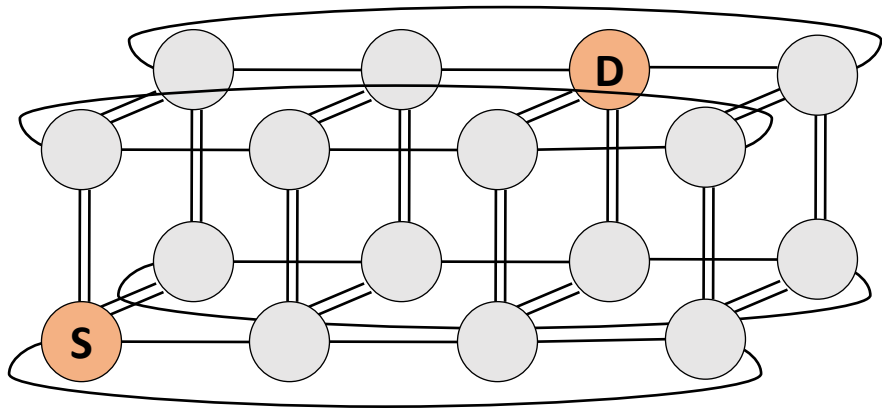
How many hops away is the furthest destination?

$k_1/2 + \dots + k_n/2 = 4$ hops (i.e., network diameter)

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

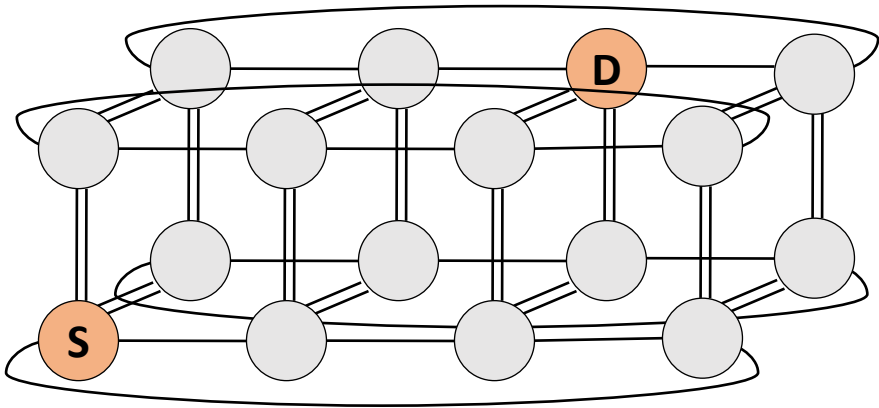


How many directions can we take?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



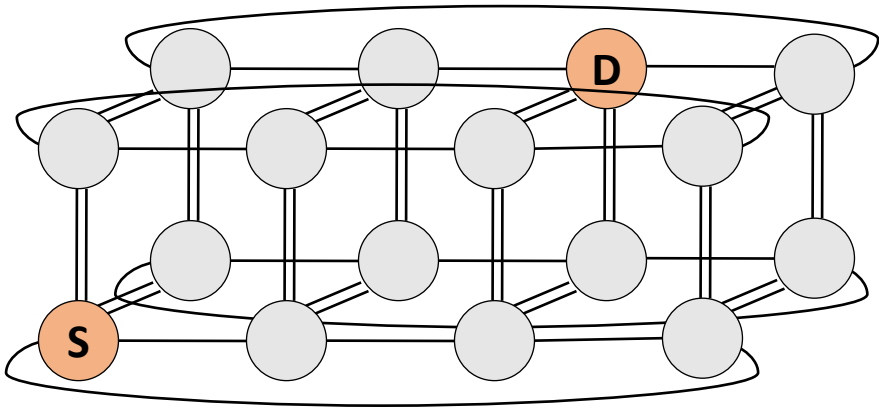
How many directions can we take?

$2^n = 8$ directions

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

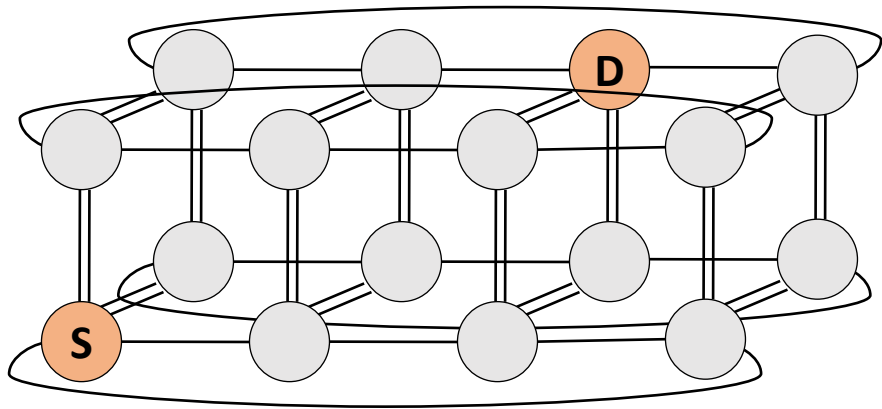


How many paths per direction do we have?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



How many paths per direction do we have?

12 paths per direction

$$\binom{\frac{k_1}{2} + \dots + \frac{k_n}{2}}{\frac{k_1}{2}, \dots, \frac{k_n}{2}} = \frac{\left(\frac{k_1}{2} + \dots + \frac{k_n}{2}\right)!}{\frac{k_1}{2}! \dots \frac{k_n}{2}!} = 12$$

total hops

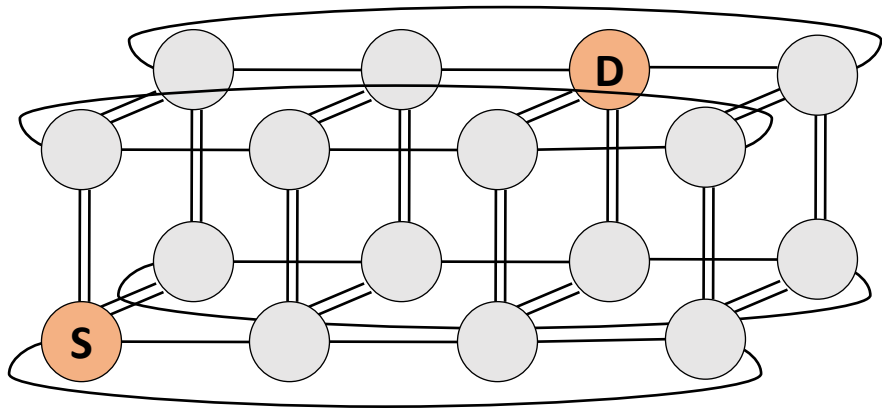
hops per dimension



Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing

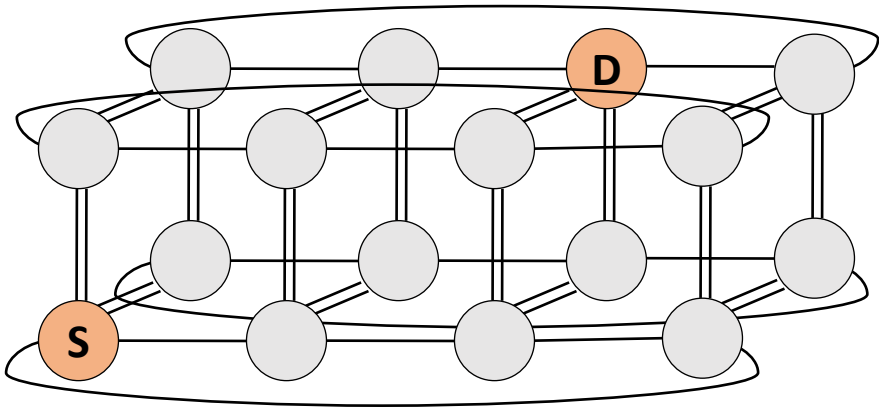


What's the maximum path diversity?

Direct Topologies – Torus

Maximum Path Diversity:

- Assume even number of routers per dimension k
- Assume shortest path routing



What's the maximum path diversity?

96 paths

$$\text{max path diversity} = 2^n \times \left(\frac{k_1}{2} + \dots + \frac{k_n}{2} \right) = 8 \times 12 = 96$$

